

COMP90077

Lecture 10

Primal-Dual and Dual-Fitting

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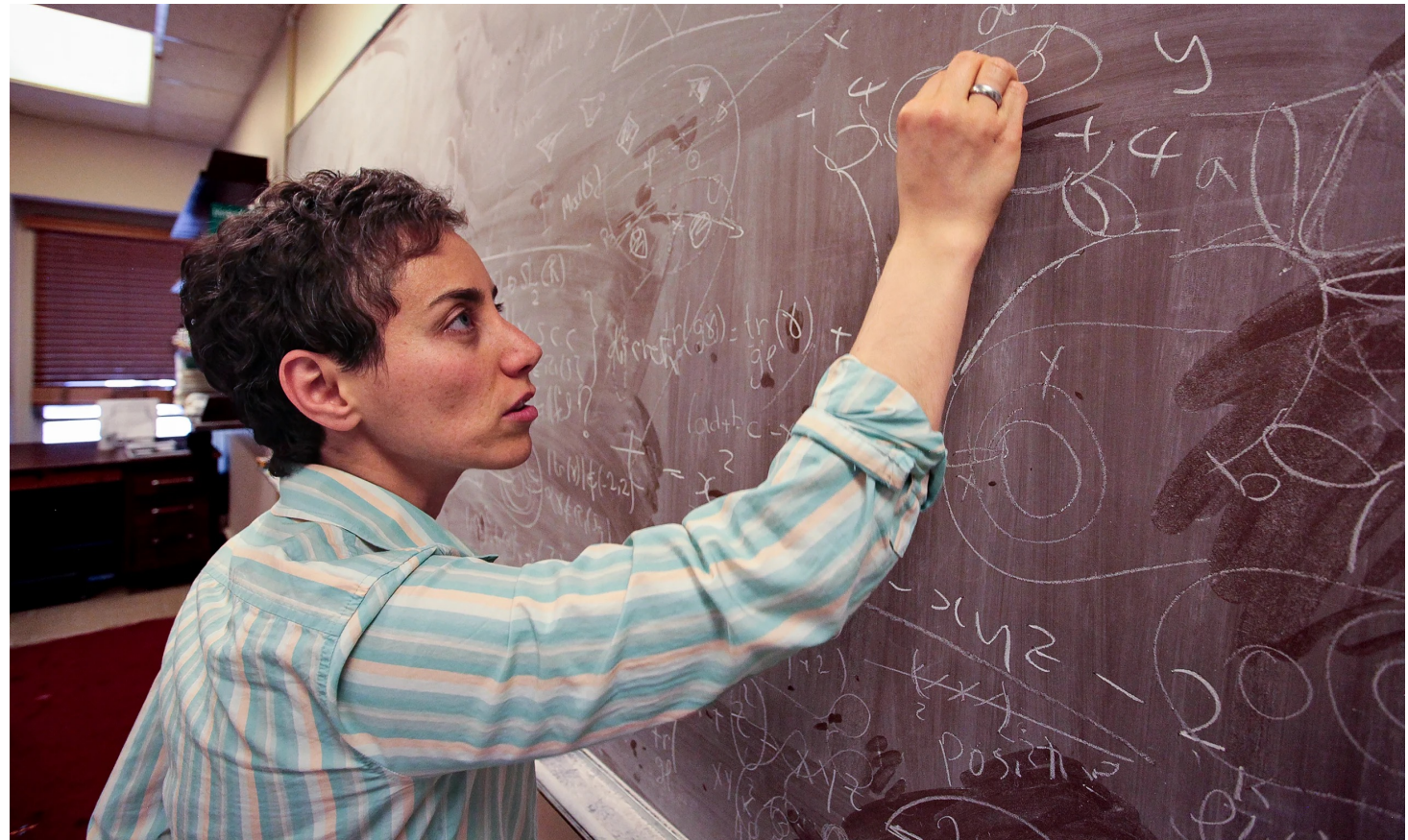
Caution

- Slides are designed as presentation aid to go along with lecture, not as study material
- Slides not designed to make sense on their own

International Women's Day in Mathematics

- May 12
- Honor achievement of women in maths
- Foster open and inclusive community
- Birthday of Miryam Mirzakhani
- First female Fields Medalist

Image: Quanta Magazine



Alison Doig – “Grandmother” of Optimization

- From Melbourne!
- Co-invented Branch-and-Bound
- Retired in 1994
- Tutored Stats in UniMelb till 90

Image: Faculty of Science



Today

- Last time: LP Duality
- 2 dual-based approaches:
 - Primal-Dual
 - Set Cover f -approximation
 - Shortest Paths
 - Minimum Spanning Trees
 - Dual-Fitting
 - Greedy for Weighted Set Cover
- Assignment 2 out later today
 - Due: in 2 weeks

Greedy for Weighted Set Cover

Sets $s_1, \dots, s_m \subseteq U$, $|U| = n$

Weights $w_1, \dots, w_m$

$$\min \sum_i x_i \cdot w_i$$

$$\text{s.t. } \sum_{s_i \ni e} x_i \geq 1 \quad \forall e \in U \quad (y_e)$$

$$\max \sum_e y_e$$

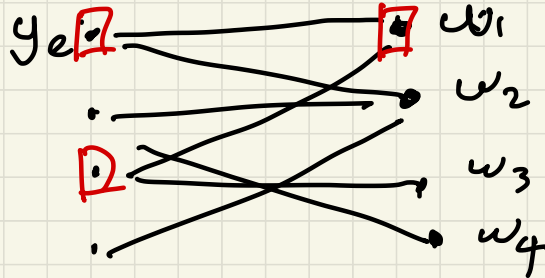
$$\text{s.t. } \sum_{e \in s_i} y_e \leq w_i \quad \forall i \quad (x_i)$$

"Economic Interpretation"

Payment of y_e
for element e

Elements

Sets



Greedy Algorithm

$$I \leftarrow \emptyset$$

$$C \leftarrow \emptyset$$

while $C \neq U$:

Pick i minimizing $\frac{w_i}{|S_i \setminus C|}$

Add i to I



(for every $e \in S_i \setminus C$, $y_e = \frac{w_i}{|S_i \setminus C|}$)

Analysis

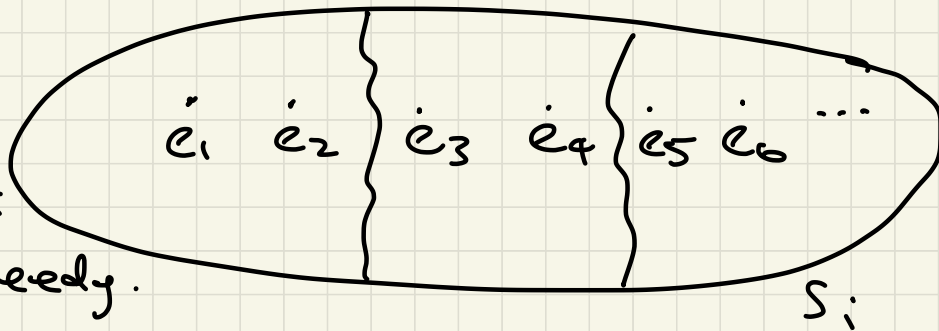
① (Dual pays for primal) $\sum_{i \in I} w_i = \sum_e y_e$

② (Dual violation) $\sum_{e \in S_i} y_e \leq H_n \cdot w_i \quad \forall i$
 $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} = \Theta(\log n)$

$$\sum_{e \in S_i} y_e \leq H_n \cdot w_i \quad \forall i$$

Pf Fix S_i

e_k is k -th elt of S_i
to be covered by greedy.



$$y_{e_k} \leq \frac{w_i}{|S_i| - k + 1}$$

In iteration e_k was covered, S_i is an option.

$$|S_i \cap C| \leq k-1 \Rightarrow |S_i \setminus C| \geq |S_i| - k + 1$$

$$y_{e_k} \leq \frac{w_i}{|S_i| - k + 1}$$

$$\sum_{k=1}^{|S_i|} y_{e_k} \leq \left(\sum_{k=1}^{|S_i|} \frac{1}{|S_i| - k + 1} \right) \cdot w_i = \left(\sum_{k=1}^{|S_i|} \frac{1}{k} \right) \cdot w_i \leq H_n \cdot w_i$$

Primal-Dual: The Tao of Algorithms

- First time that algorithms made sense to me
- Unifies shortest-path and minimum spanning trees
- Today, I always first try writing an LP and its dual



Primal

- Integral primal solution X
- Cost of $X \leq \alpha$ value of y
- X is α -approximation 🎉

Dual

- Feasible dual solution y
- value of y
 $\leq$ cost of opt fractional primal
 $\leq$ cost of opt integral primal

Primal-Dual:
Construct X and y in tandem

Complementary Slackness Conditions

- x^*, y^* are optimal for primal and dual LPs if and only if they satisfy CS conditions
- (Primal CS) Positive primal variables correspond to tight dual constraints
- (Dual CS) Positive dual variables correspond to tight primal constraints

But first, an example

Example

maximize $x_1 + 6x_2$

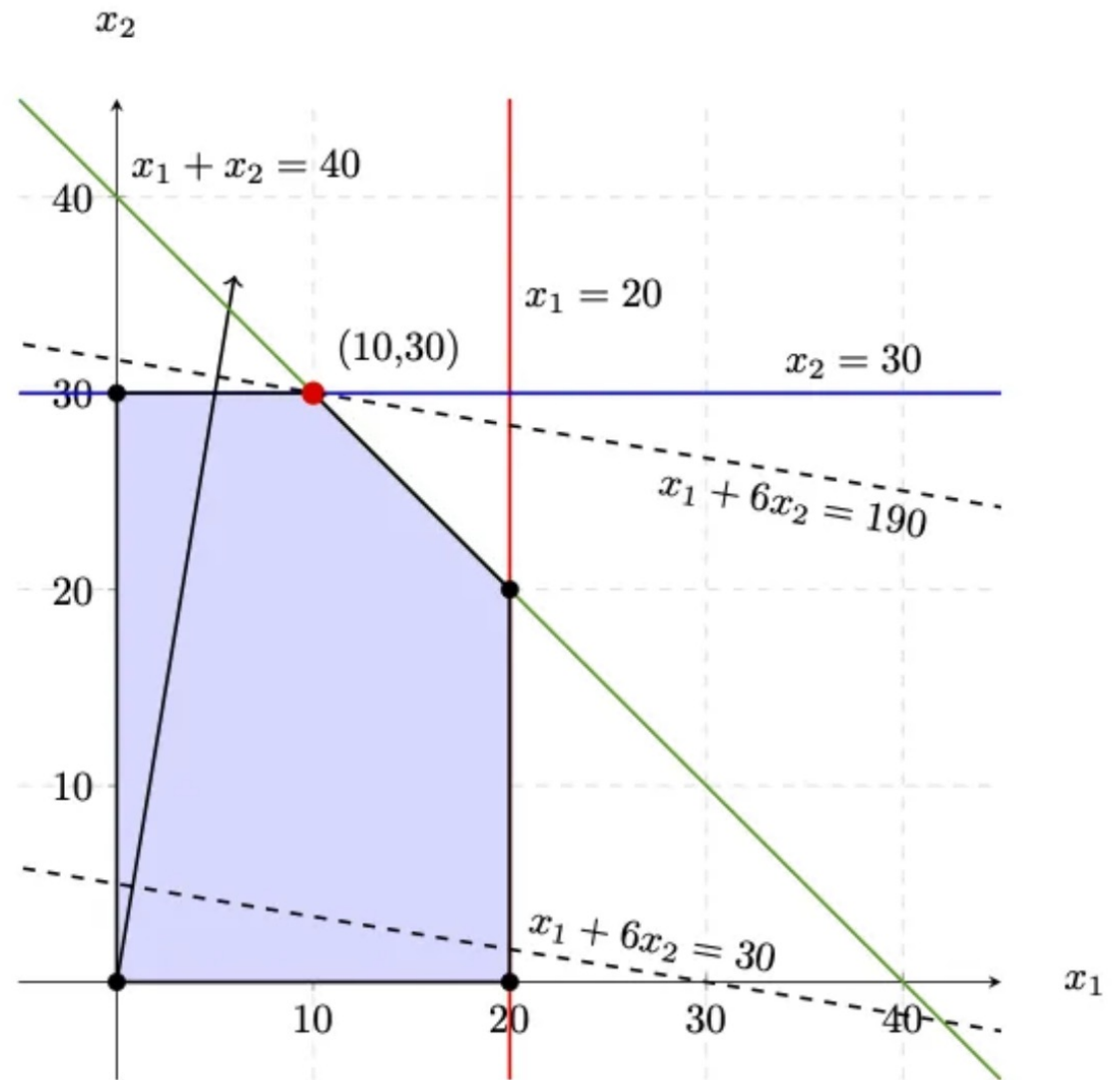
subject to $x_1 \leq 20$ (y_1)
 $x_2 \leq 30$ (y_2)
 $x_1 + x_2 \leq 40$ (y_3)
 $x \geq 0$

$$x_1^* = 10, x_2^* = 30$$

minimize $20y_1 + 30y_2 + 40y_3$

subject to $y_1 + y_3 \geq 1$
 $y_2 + y_3 \geq 6$
 $y \geq 0$

$$y_1^* = 0, y_2^* = 5, y_3^* = 1$$

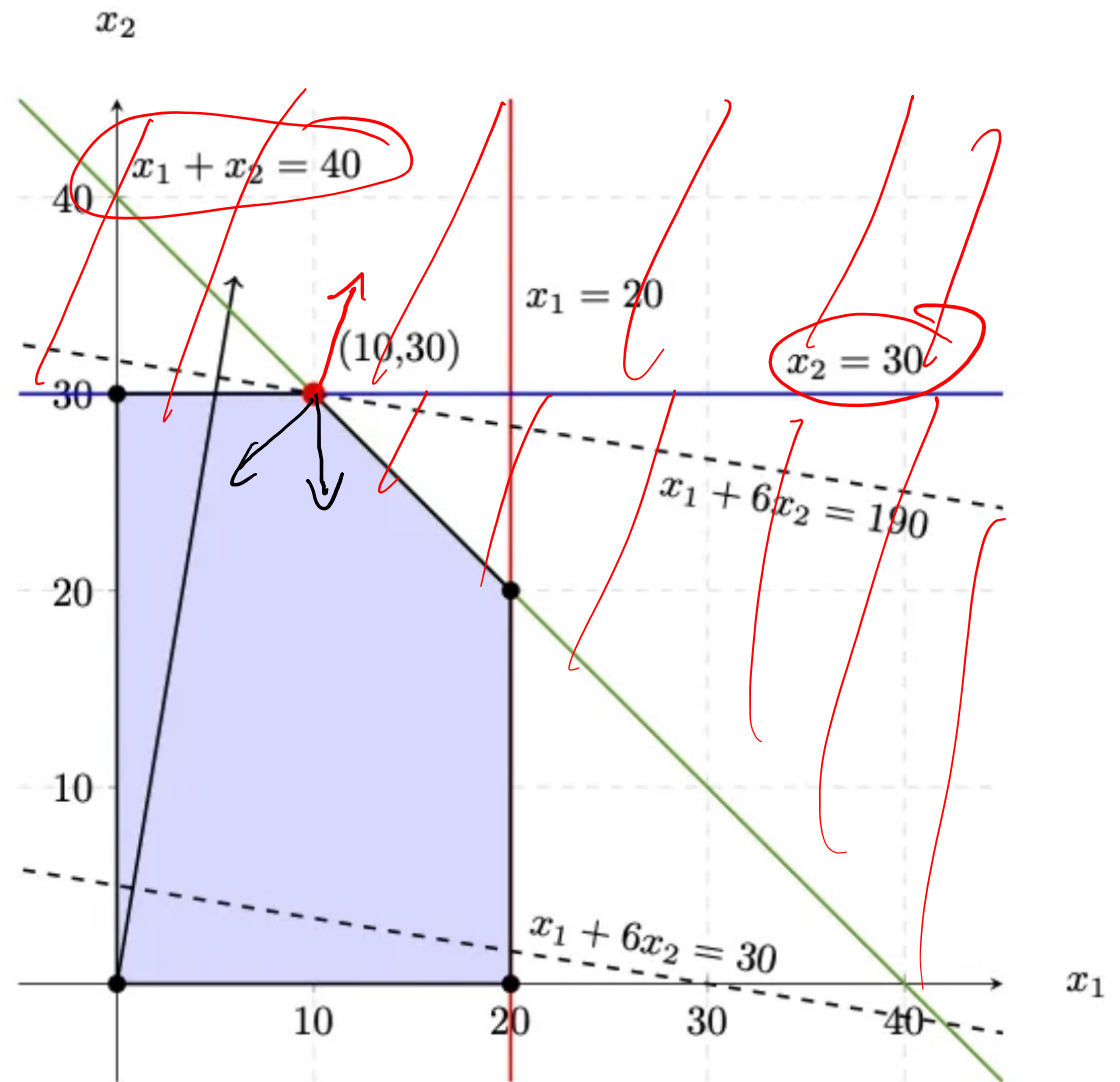


$$\begin{array}{ll}
 \text{maximize} & x_1 + 6x_2 \\
 \text{subject to} & x_1 \leq 20 \\
 & x_2 \leq 30 \\
 & x_1 + x_2 \leq 40 \\
 & x \geq 0
 \end{array}$$

$$x_1^* = 10, x_2^* = 30$$

$$y_1^* = 0, y_2^* = 5, y_3^* = 1$$

- Dual solution can be used to certify optimality of primal solution
- Primal solution optimal iff tight constraints rule out improving direction



Complementary Slackness Conditions

- x^*, y^* are optimal for primal and dual LPs if and only if they satisfy CS conditions
- (Primal CS) If $x_j^* > 0$, then $\sum_{1 \leq i \leq m} b_{ij} y_i^* = a_j$
- (Dual CS) If $y_i^* > 0$, then $\sum_{1 \leq j \leq n} b_{ij} x_j^* = c_i$

$$\begin{array}{ll}
 \text{maximize} & \sum_{1 \leq j \leq n} a_j x_j \\
 \text{subject to} & \sum_{1 \leq j \leq n} b_{ij} x_j \leq c_i \quad \forall 1 \leq i \leq m \\
 & x \geq 0
 \end{array}$$

$$\begin{array}{ll}
 \text{minimize} & \sum_{1 \leq i \leq m} c_i y_i \\
 \text{subject to} & \sum_{1 \leq i \leq m} b_{ij} y_i \geq a_j \quad \forall 1 \leq j \leq n \\
 & y \geq 0
 \end{array}$$

Complementary Slackness Conditions

If x^*, y^* are feasible and satisfy CS conditions, then they are optimal

- (Primal CS) If $x_j^* > 0$, then $\sum_{1 \leq i \leq n} b_{ij} y_i^* = a_j$
- (Dual CS) If $y_i^* > 0$, then $\sum_{1 \leq j \leq n} b_{ij} x_j^* = c_i$

Proof.

$$\begin{aligned}
 \sum_{1 \leq j \leq n} a_j x_j^* &= \sum_{j: x_j^* > 0} a_j x_j^* \\
 &= \sum_{j: x_j^* > 0} \left(\sum_{i: y_i^* > 0} b_{ij} y_i^* \right) x_j^* = \sum_{i: y_i^* > 0} \left(\sum_{j: x_j^* > 0} b_{ij} x_j^* \right) y_i^* \\
 &= \sum_{1 \leq i \leq m} c_i y_i^* \quad \square
 \end{aligned}$$

Weighted Set Cover: f-approximation via Primal-Dual

Given:

- set U of n elements,
- m sets $S_1, \dots, S_m \subseteq U$ with costs $w_1, \dots, w_m$
- Find collection of sets covering all elements with cheapest total cost

$$\begin{array}{ll} \text{minimize} & \sum_i w_i x_i \\ \text{subject to} & \sum_{i: S_i \ni e} x_i \geq 1 \quad \forall e \in U \\ & x \geq 0 \end{array}$$

$$\begin{array}{ll} \text{maximize} & \sum_e y_e \\ \text{subject to} & \sum_{e: S_i \ni e} y_e \leq w_i \quad \forall i \in [m] \\ & y \geq 0 \end{array}$$

Weighted Set Cover: Complementary Slackness

$$\begin{array}{ll} \text{minimize} & \sum_i w_i x_i \\ \text{subject to} & \sum_{i: S_i \ni e} x_i \geq 1 \quad \forall e \in U \\ & x \geq 0 \end{array}$$

$$\begin{array}{ll} \text{maximize} & \sum_e y_e \\ \text{subject to} & \sum_{e \in S_i} y_e \leq w_i \quad \forall i \in [m] \\ & y \geq 0 \end{array}$$

Integral primal X and dual solution y satisfying CS means:

- (Primal CS) If $X_i = 1$, then $\sum_{e \in S_i} y_e = w_i$ (dual pays for primal)
- (Dual CS) If $y_e > 0$, then $\sum_{i: S_i \ni e} X_i = 1$ (dual not overcharged)

$$\sum_i w_i x_i = \sum_i \left(\sum_{e \in S_i} y_e \right) x_i = \sum_e \left(\sum_{i: S_i \ni e} x_i \right) y_e = \sum_e y_e$$

Weighted Set Cover: Complementary Slackness

$$\begin{array}{ll} \text{minimize} & \sum_i w_i x_i \\ \text{subject to} & \sum_{i: S_i \ni e} x_i \geq 1 \quad \forall e \in U \\ & x \geq 0 \end{array}$$

$$\begin{array}{ll} \text{maximize} & \sum_e y_e \\ \text{subject to} & \sum_{e \in S_i} y_e \leq w_i \quad \forall i \in [m] \\ & y \geq 0 \end{array}$$

Integral primal X and dual solution y satisfying CS means:

- (Primal CS) If $X_i = 1$, then $\sum_{e \in S_i} y_e = w_i$ (dual pays for primal)
- (α -approx. Dual CS) If $y_e > 0$, then $\sum_{i: S_i \ni e} X_i \leq \alpha$ (dual α -overcharged)

$$\sum_i w_i X_i = \sum_i \left(\sum_{e \in S_i} y_e \right) X_i = \sum_e \left(\sum_{i: S_i \ni e} X_i \right) y_e \leq \alpha \sum_e y_e$$

Weighted Set Cover: Primal-Dual

Primal-Dual Algorithm:

- Initialize $X, y = 0$
- While there is an uncovered element e ,
 - raise y_e until $\sum_{e \in S_i} y_e = w_i$ for some i
 - $X_i = 1$ (buy set S_i)

Analysis:

- (Primal CS) If $X_i = 1$, then $\sum_{e \in S_i} y_e = w_i$
- (f -approx. Dual CS) If $y_e > 0$, then $\sum_{i: S_i \ni e} X_i \leq f$
- 🎉

$$\begin{aligned}
 &\text{minimize} && \sum_i w_i x_i \\
 &\text{subject to} && \sum_{i: S_i \ni e} x_i \geq 1 && \forall e \in U \\
 &&& x \geq 0 \\
 \\
 &\text{maximize} && \sum_e y_e \\
 &\text{subject to} && \sum_{e \in S_i} y_e \leq w_i && \forall i \in [m] \\
 &&& y \geq 0
 \end{aligned}$$

General Primal-Dual for Covering

Covering

$$\begin{array}{ll} \min_{S \subseteq E} & \sum_{e \in S} c_e \\ \text{subject to} & S \in \mathcal{F} \end{array}$$

where if S feasible, then S' is also feasible for every $S' \supseteq S$.

- Examples: vertex cover, set cover, MST, min-weight basis (matroid)
- Problems that can be reduced to Set Cover
 - possibly with exponentially many sets and elements

Packing

$$\begin{array}{ll} \max_{S \subseteq E} & \sum_{e \in S} w_e \\ \text{subject to} & S \in \mathcal{F} \end{array}$$

where if S feasible, then S' is also feasible for every $S' \subseteq S$.

- Examples: Knapsack, matching
- Problems that can be reduced to Max-Weight Independent Set (Independence Systems)

Covering LP

$$\begin{array}{ll}\min_{S \subseteq E} & \sum_{e \in S} c_e \\ \text{subject to} & S \in \mathcal{F}\end{array}$$

where if S feasible, then S' is also feasible for every $S' \supseteq S$.

$$\begin{array}{ll}\text{minimize} & \sum_{1 \leq i \leq n} c_i x_i \\ \text{subject to} & \sum_{1 \leq i \leq n} b_{ij} x_i \geq d_j \quad \forall 1 \leq j \leq m \\ & x \geq 0\end{array}$$

$$c_j, b_{ij}, d_j \geq 0$$

Packing LP

$$\begin{array}{ll}\max_{S \subseteq E} & \sum_{e \in S} d_e \\ \text{subject to} & S \in \mathcal{F}\end{array}$$

where if S feasible, then S' is also feasible for every $S' \subseteq S$.

$$\begin{array}{ll}\text{maximize} & \sum_{1 \leq j \leq n} d_j y_j \\ \text{subject to} & \sum_{1 \leq j \leq n} b_{ij} y_j \leq c_i \quad \forall 1 \leq i \leq n \\ & y \geq 0\end{array}$$

$$c_j, b_{ij}, d_j \geq 0$$

Basic Primal-Dual for Covering Problems

Primal-Dual Algorithm:

- Initialize $X, y = 0$
- While some covering constraint violated,
 - raise its dual variable until dual constraint tight
 - Set its primal variable to 1

Analysis:

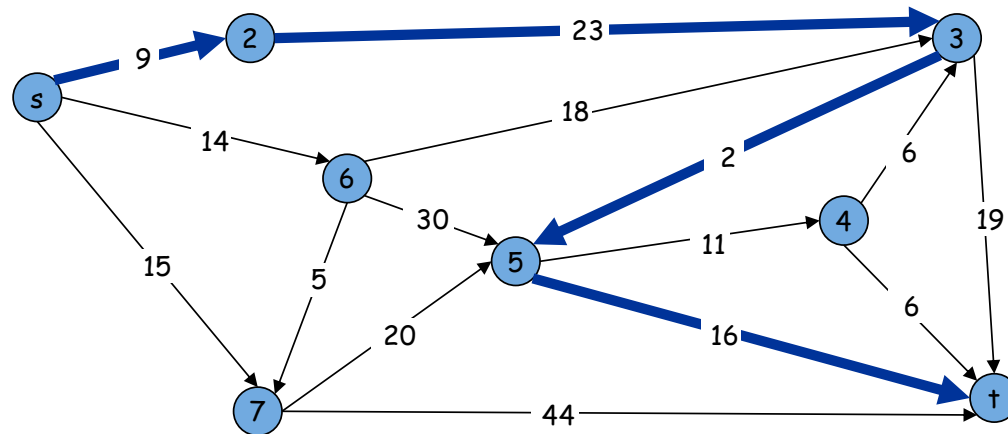
- (Primal CS) If $X_i = 1$, then $\sum_{1 \leq j \leq n} y_j = w_i$
- (α -approx. Dual CS) If $y_e > 0$, then $\sum_{1 \leq i \leq n} X_i \leq \alpha d_j$
- Conclude algorithm is α -approximation
- 🎉

$$\begin{array}{ll}\text{minimize} & \sum_{1 \leq i \leq n} c_i x_i \\ \text{subject to} & \sum_{1 \leq i \leq n} b_{ij} x_i \geq d_j \quad \forall 1 \leq j \leq m \\ & x \geq 0\end{array}$$

$$\begin{array}{ll}\text{maximize} & \sum_{1 \leq j \leq m} d_j y_j \\ \text{subject to} & \sum_{1 \leq j \leq m} b_{ij} y_j \leq c_i \quad \forall 1 \leq i \leq n \\ & y \geq 0\end{array}$$

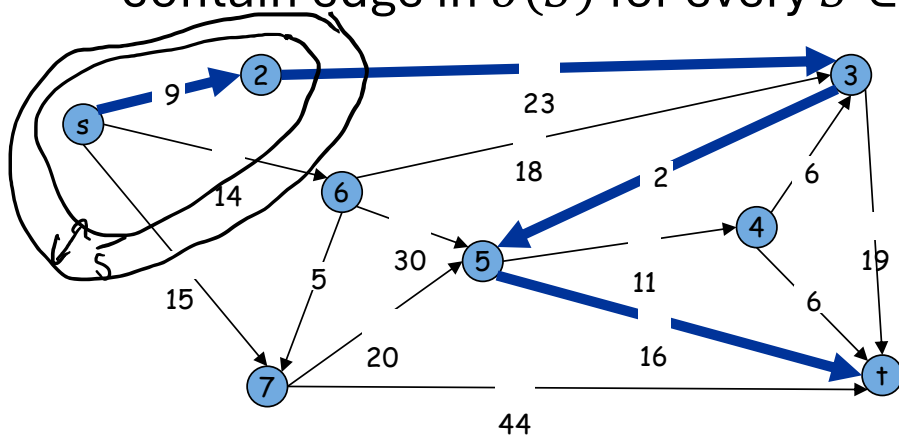
Shortest (s,t)-Path in Directed Graphs

- Given directed graph $G = (V, E)$ with edge costs c_e , and vertices $s, t \in V$, find shortest (s, t) -path.
- How to write LP?



Primal and dual LPs

- Given directed graph $G = (V, E)$ with edge costs c_e , and vertices $s, t \in V$, find shortest (s, t) -path.
- Let $\mathcal{C}$ = set of (s, t) -cuts, i.e. $S \subseteq V$ such that $s \in S, t \notin S$
- Let $\delta(S) = \{(u, v) : u \in S, v \notin S\}$, edges going out from S
- An (s, t) -path must cross $\mathcal{C}$, i.e. contain edge in $\delta(S)$ for every $S \in \mathcal{C}$



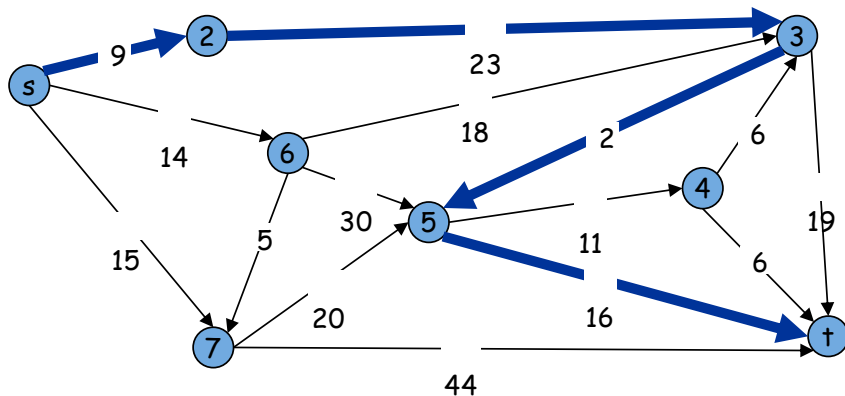
$$\begin{aligned}
 &\text{minimize} && \sum_{e \in E} c_e x_e \\
 &\text{subject to} && \sum_{e \in \delta(S)} x_e \geq 1 \quad \forall S \in \mathcal{C} \\
 &&& x \geq 0
 \end{aligned}$$

$$\begin{aligned}
 &\text{maximize} && \sum_{S \in \mathcal{C}} y_S \\
 &\text{subject to} && \sum_{S: e \in \delta(S)} y_S \leq c_e \quad \forall e \in E \\
 &&& y \geq 0
 \end{aligned}$$

Intuition: Dual = “moat” packing

Primal-Dual Algorithm

- Initialize $F = \emptyset$ and $y = 0$
- While there exists violated cut:
 - which dual variables to raise?



$$\begin{aligned}
 &\text{minimize} && \sum_{e \in E} c_e x_e \\
 &\text{subject to} && \sum_{e \in \delta(S)} x_e \geq 1 \quad \forall S \in \mathcal{C} \\
 &&& x \geq 0
 \end{aligned}$$

$$\begin{aligned}
 &\text{maximize} && \sum_{S \in \mathcal{C}} y_S \\
 &\text{subject to} && \sum_{S: e \in \delta(S)} y_S \leq c_e \quad \forall e \in E \\
 &&& y \geq 0
 \end{aligned}$$

Intuition: Dual = “moat” packing

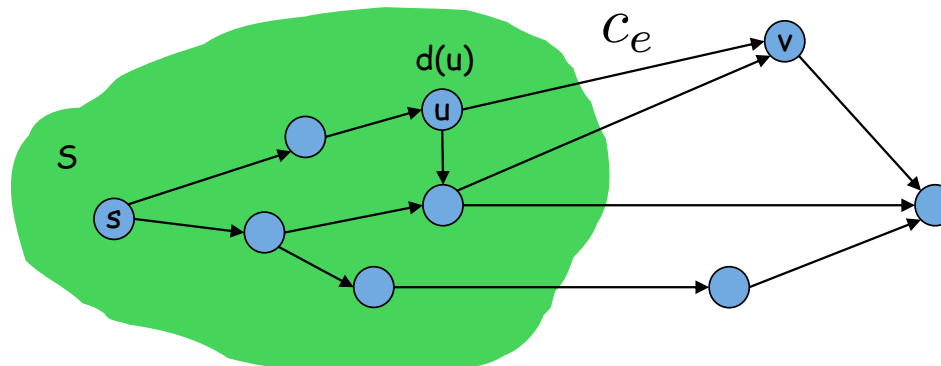
Dijkstra's Algorithm

- Compute shortest path distances to every vertex.
- Dijkstra's algorithm.
 - Maintain a set of **explored nodes** S and the shortest path tree T rooted at s spanning S
 - Initialize $S = \{s\}$, $d(s) = 0$, $T = \{\}$
 - Repeatedly choose unexplored node v and edge e which minimizes
$$\pi(v) = \min_{e=(u,v): u \in S} d(u) + c_e$$
add v to S , e to T , and set $d(v) = \pi(v)$.
 - Observe: s - v path in T has length $d(v)$

← shortest path to some u in explored part, followed by a single edge (u, v)

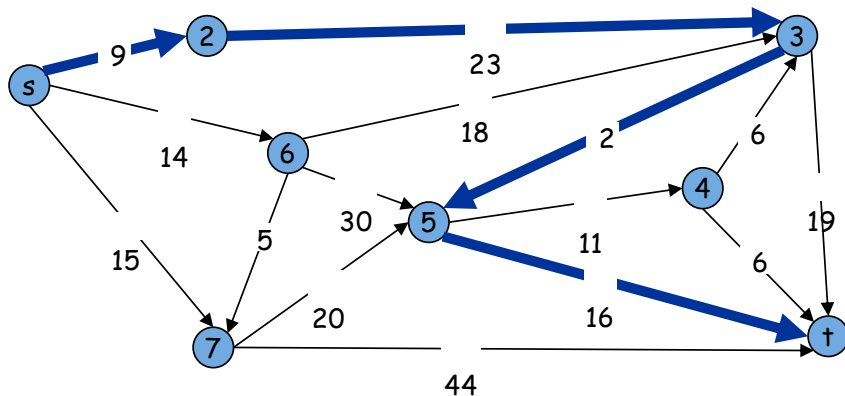
Idea: Mimic shortest path tree growing process

Equivalent to BFS if edge costs are 1



Primal-Dual Algorithm: Moat-Growing

- Initialize $T = \emptyset$, $y = 0$, $S = \{s\}$
- While $t \notin S$:
 - Raise y_S until some edge (u, v) tight
 - Add (u, v) to T



$$\begin{aligned}
 &\text{minimize} && \sum_{e \in E} c_e x_e \\
 &\text{subject to} && \sum_{e \in \delta(S)} x_e \geq 1 \quad \forall S \in \mathcal{C} \\
 &&& x \geq 0
 \end{aligned}$$

$$\begin{aligned}
 &\text{maximize} && \sum_{S \in \mathcal{C}} y_S \\
 &\text{subject to} && \sum_{S: e \in \delta(S)} y_S \leq c_e \quad \forall e \in E \\
 &&& y \geq 0
 \end{aligned}$$

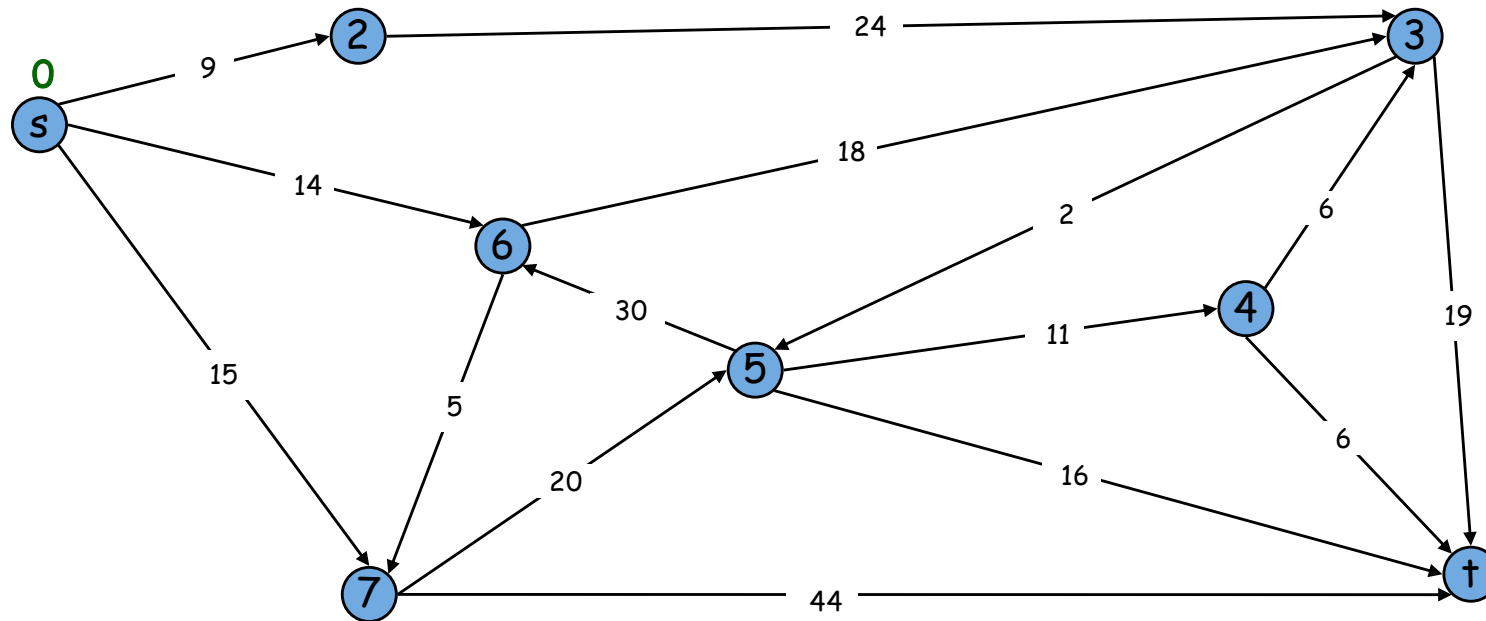
Intuition: Dual = “moat” packing

Illustration of Moat-Growing Process

Moat-Growing Process

$S = \{s\}$

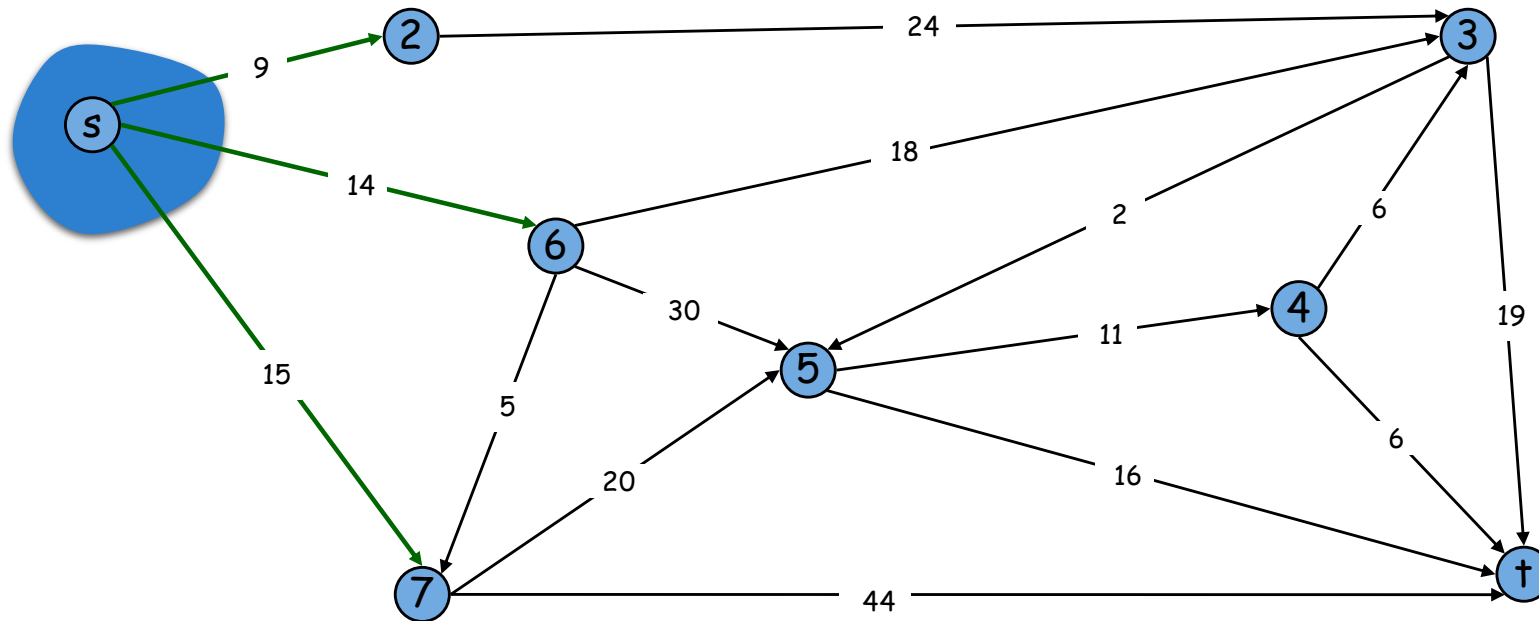
$T = \{\}$



Moat-Growing Process

$S = \{s\}$
 $T = \{\}$

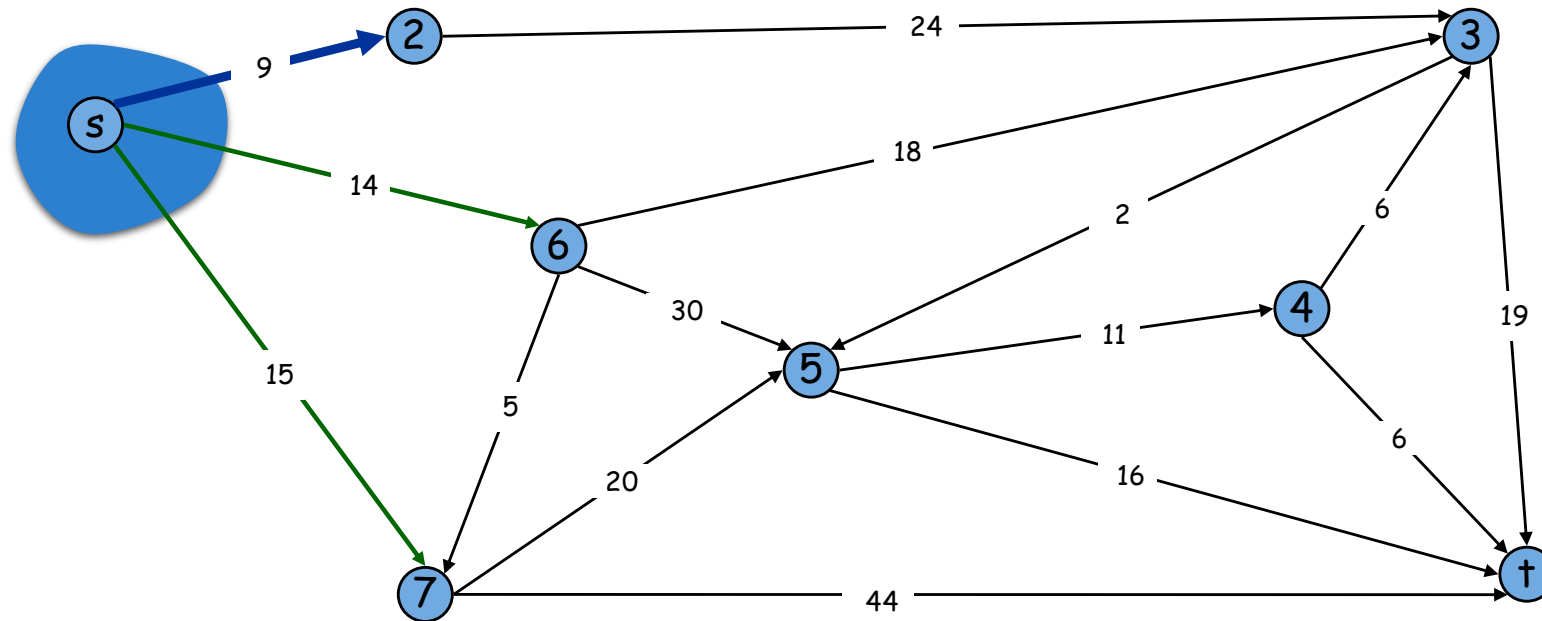
$y_{\{s\}} = ?$



Moat-Growing Process

$S = \{s\}$
 $T = \{(s, 2)\}$

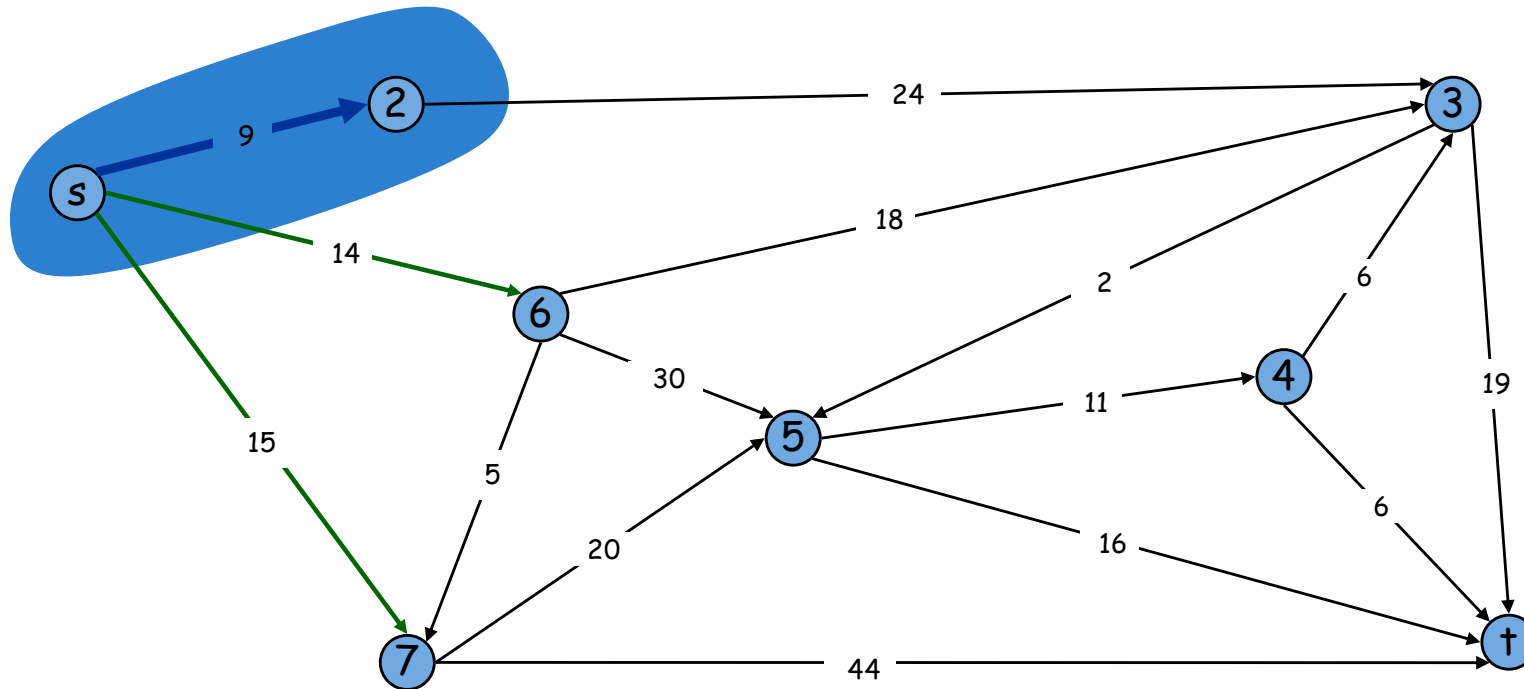
$$y_{\{s\}} = 9$$



Moat-Growing Process

$S = \{s, 2\}$
 $T = \{(s, 2)\}$

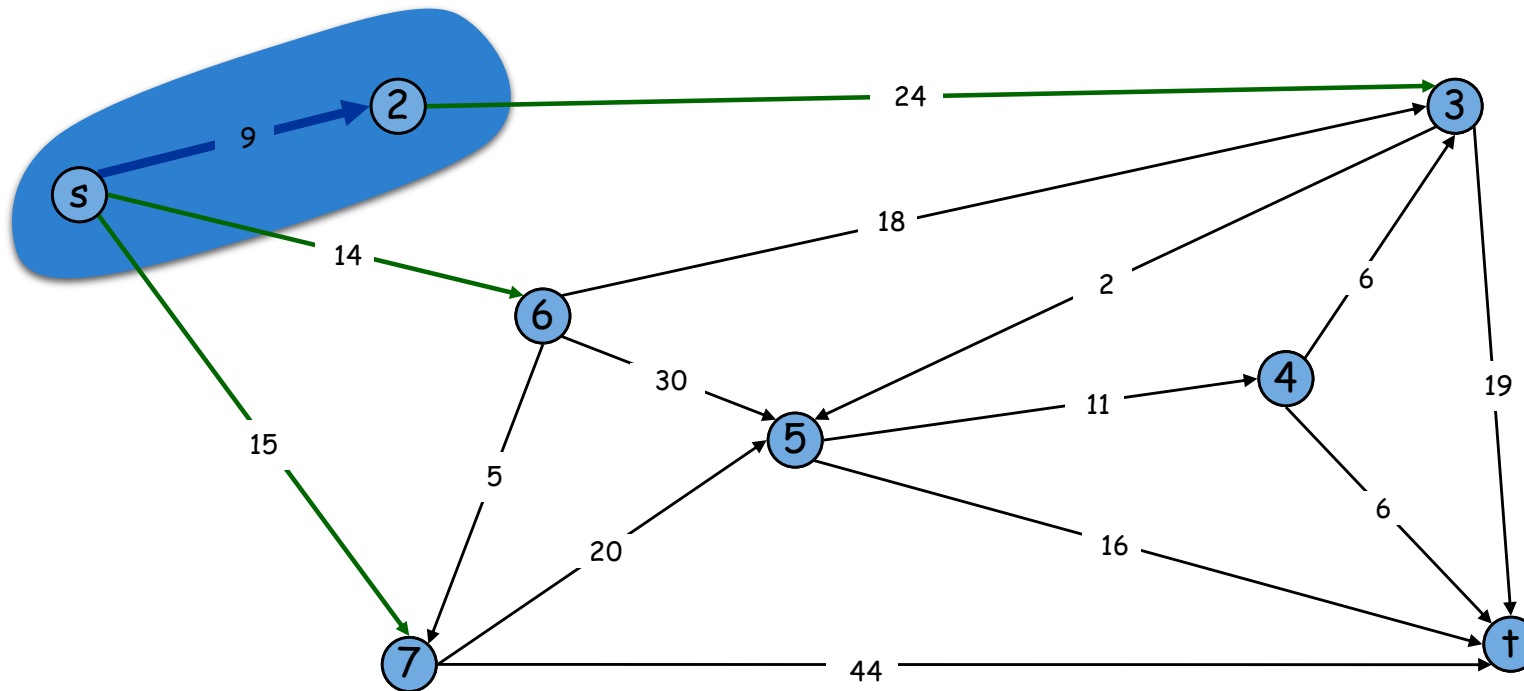
$$y_{\{s\}} = 9$$



Moat-Growing Process

$S = \{s, 2\}$
 $T = \{(s, 2)\}$

$y_{\{s\}} = 9$
 $y_{\{s, 2\}} = ?$

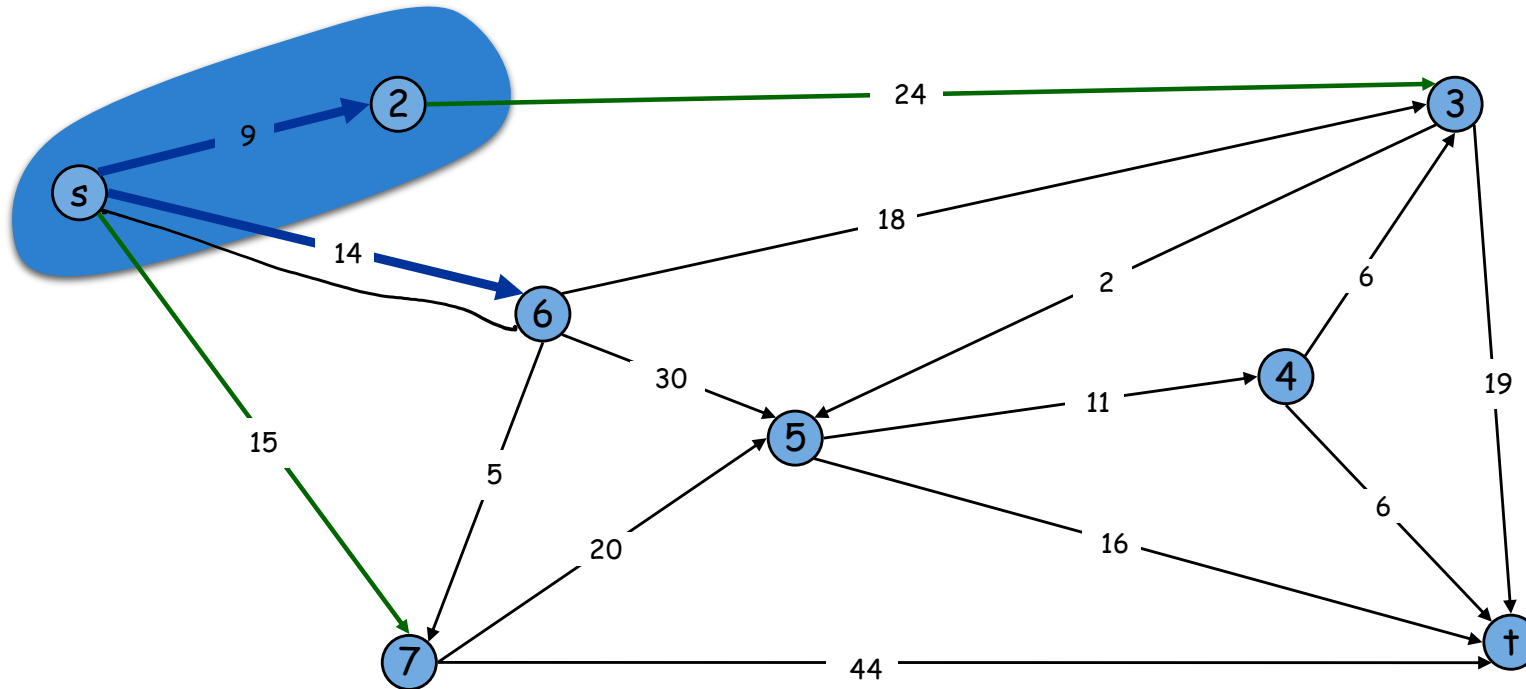


Moat-Growing Process

$$S = \{s, 2\}$$

$$T = \{(s, 2), (s, 6)\}$$

$$\boxed{y_{\{s\}} = 9}$$
$$\boxed{y_{\{s, 2\}} = 5}$$



Moat-Growing Process

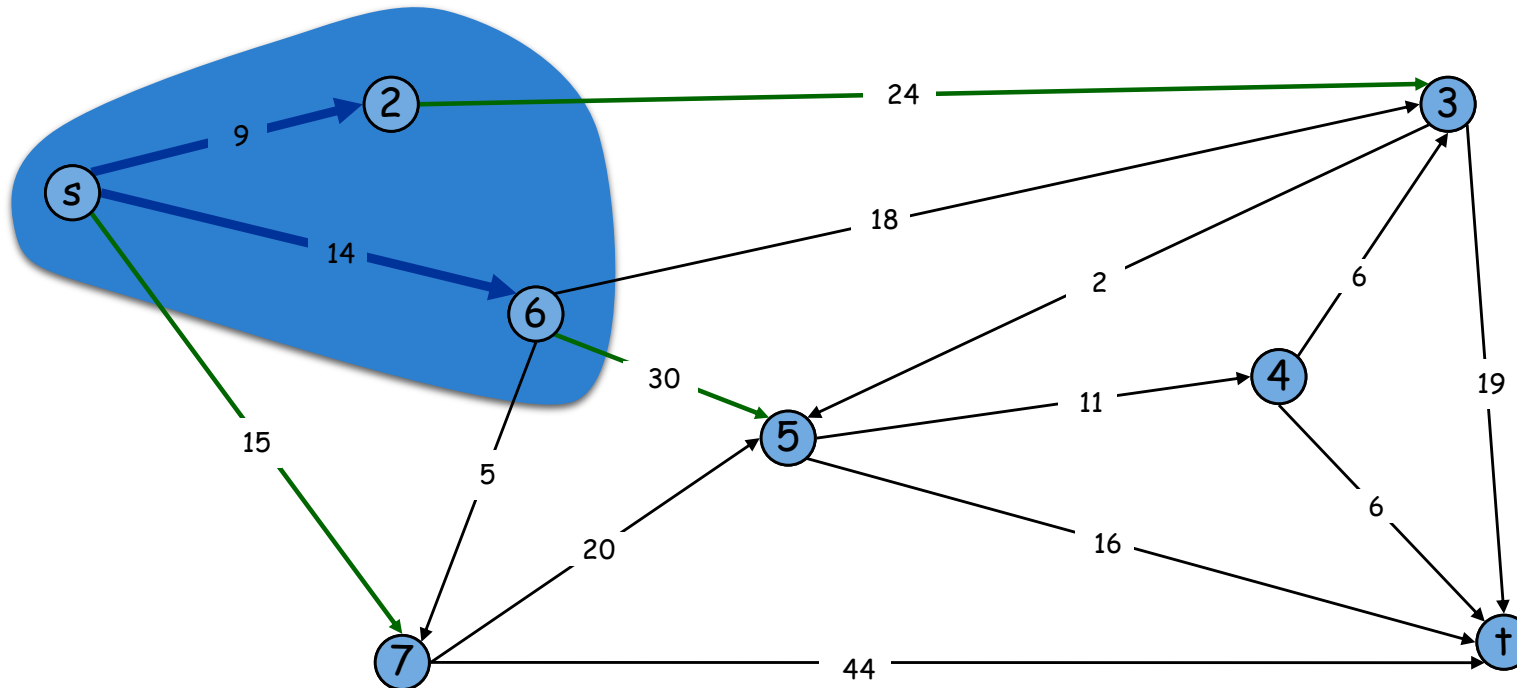
$$S = \{s, 2, 6\}$$

$$T = \{(s, 2), (s, 6)\}$$

$$y_{\{s\}} = 9$$

$$y_{\{s, 2\}} = 5$$

$$y_{\{s, 2, 6\}} = ?$$



Moat-Growing Process

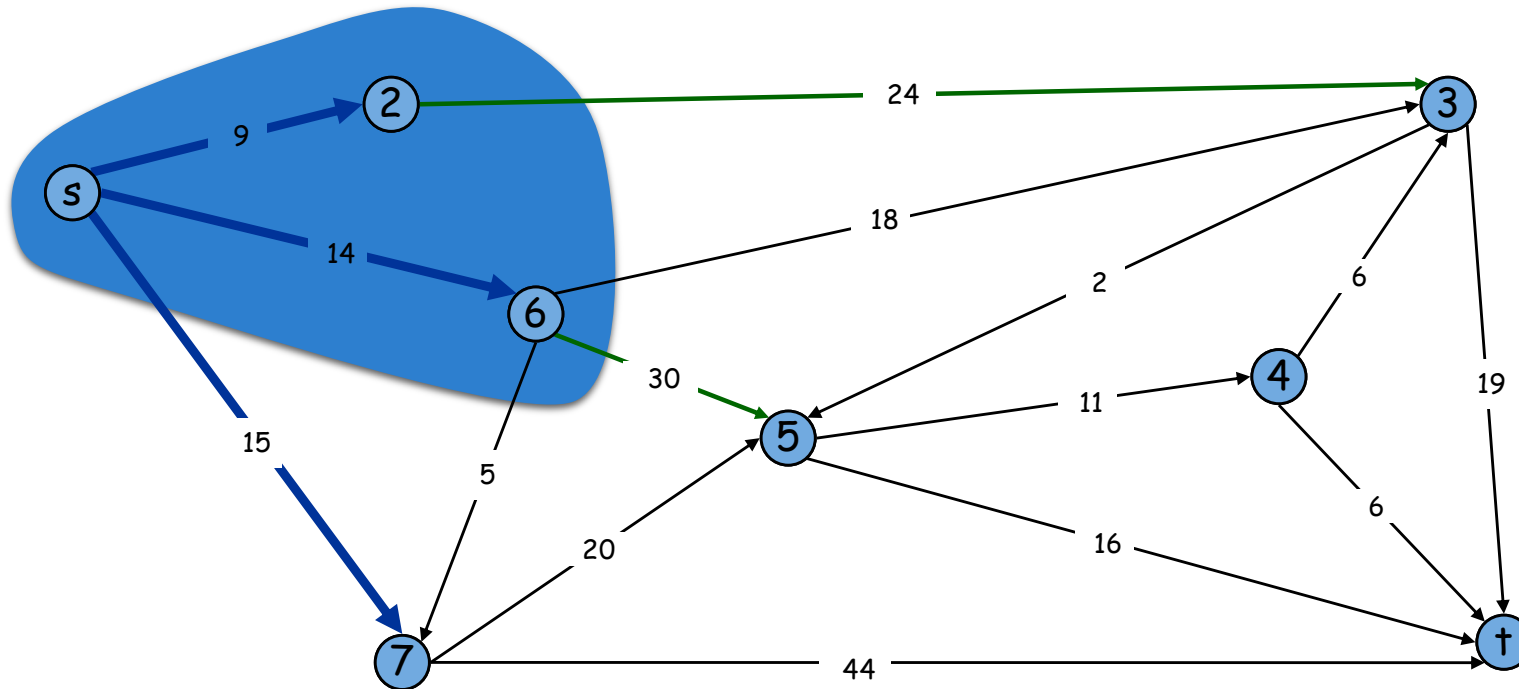
$$S = \{s, 2, 6\}$$

$$T = \{(s, 2), (s, 6), (s, 7)\}$$

$$y_{\{s\}} = 9$$

$$y_{\{s, 2\}} = 5$$

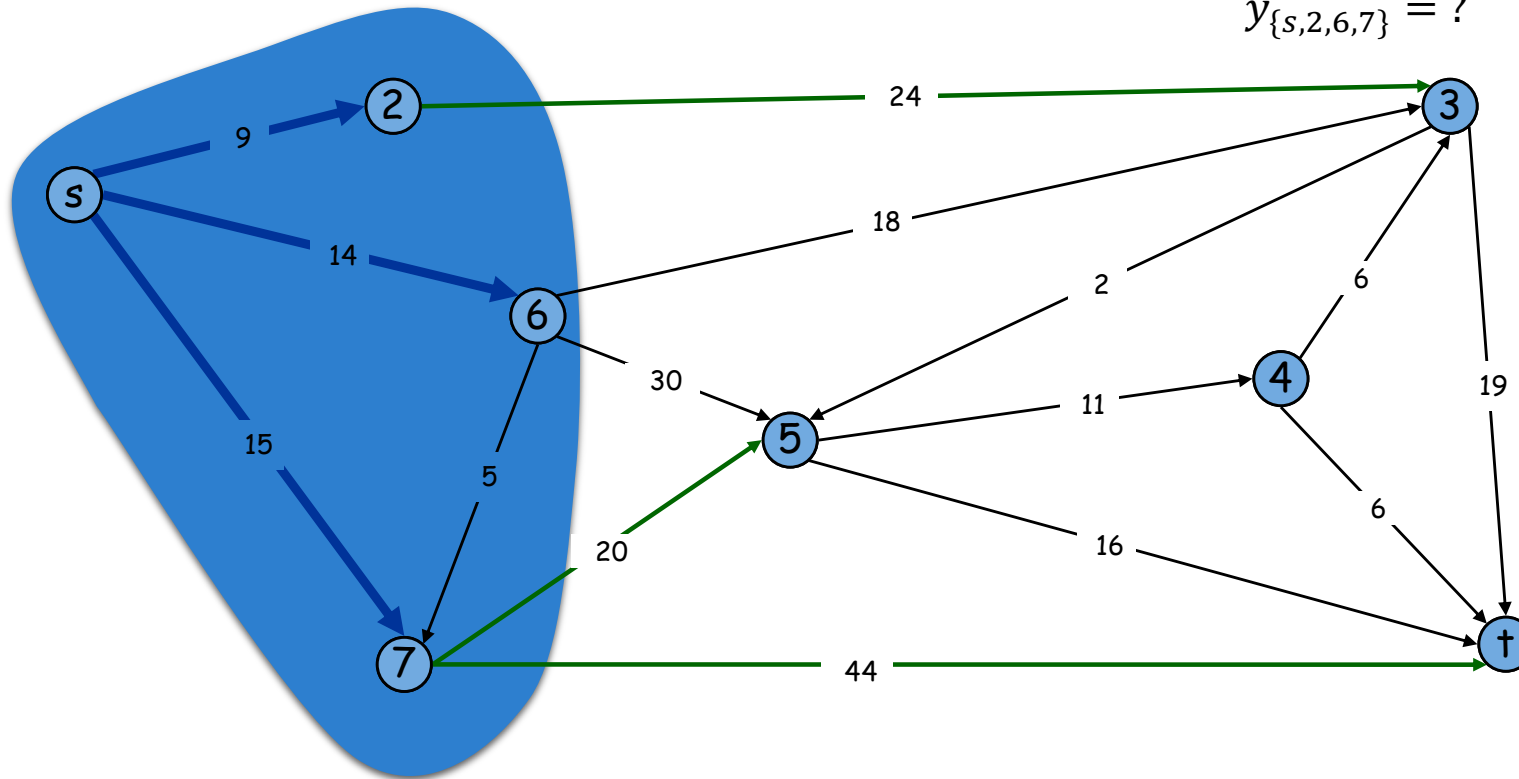
$$y_{\{s, 2, 6\}} = 1$$



Moat-Growing Process

$S = \{s, 2, 6, 7\}$
 $T = \{(s, 2), (s, 6), (s, 7)\}$

$y_{\{s\}} = 9$
 $y_{\{s, 2\}} = 5$
 $y_{\{s, 2, 6\}} = 1$
 $y_{\{s, 2, 6, 7\}} = ?$



Moat-Growing Process

$$S = \{s, 2, 6, 7\}$$

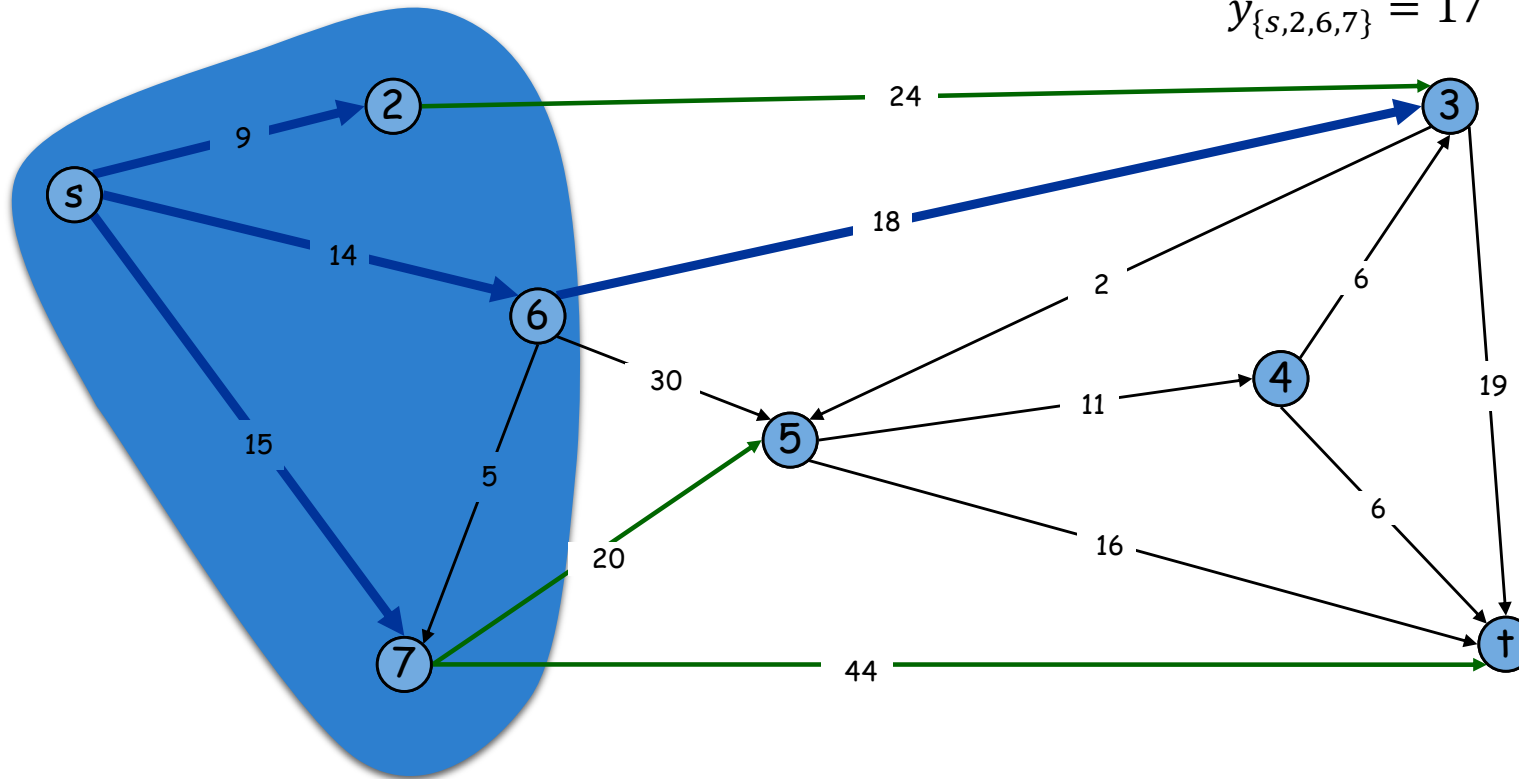
$$T = \{(s, 2), (s, 6), (s, 7), (6, 3)\}$$

$$y_{\{s\}} = 9$$

$$y_{\{s, 2\}} = 5$$

$$y_{\{s, 2, 6\}} = 1$$

$$y_{\{s, 2, 6, 7\}} = 17$$



Moat-Growing Process

$$S = \{s, 2, 6, 7, 3\}$$

$$T = \{(s, 2), (s, 6), (s, 7), (6, 3)\}$$

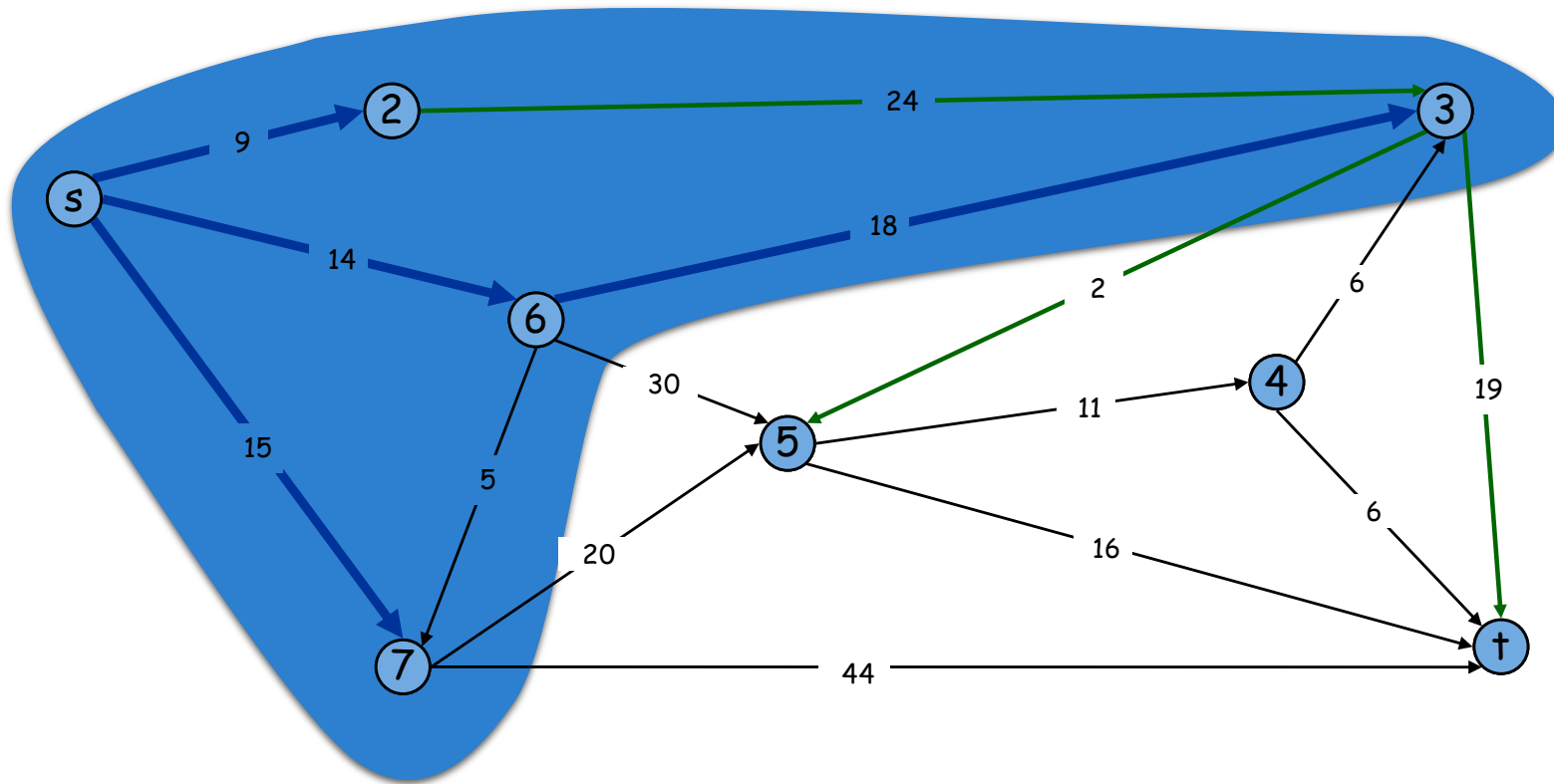
$$y_{\{s\}} = 9$$

$$y_{\{s, 2\}} = 5$$

$$y_{\{s, 2, 6\}} = 1$$

$$y_{\{s, 2, 6, 7\}} = 17$$

$$y_{\{s, 2, 6, 7, 3\}} = ?$$



Moat-Growing Process

$$S = \{s, 2, 6, 7, 3\}$$

$$T = \{(s, 2), (s, 6), (s, 7), (6, 3)\}$$

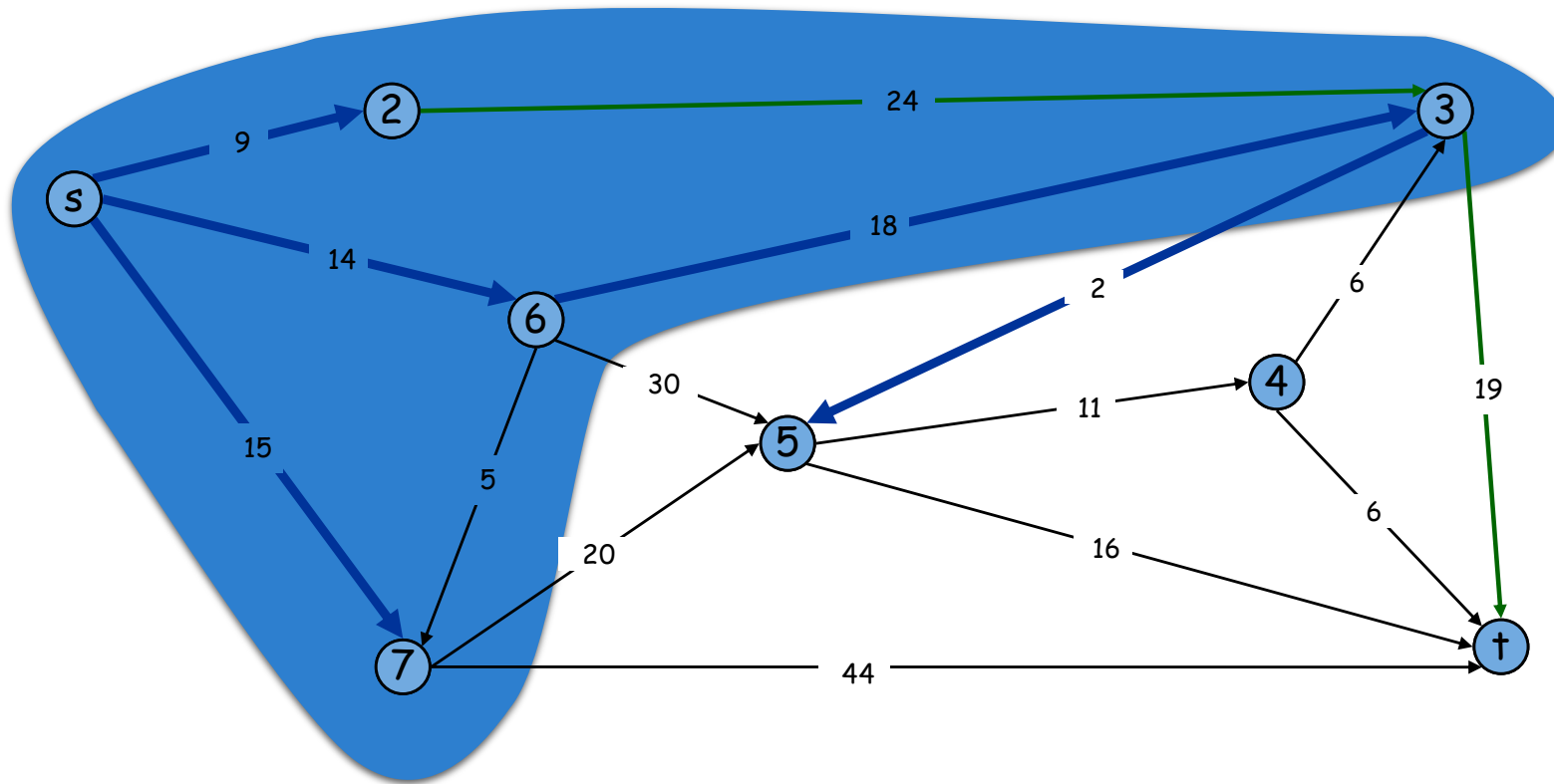
$$y_{\{s\}} = 9$$

$$y_{\{s, 2\}} = 5$$

$$y_{\{s, 2, 6\}} = 1$$

$$y_{\{s, 2, 6, 7\}} = 17$$

$$y_{\{s, 2, 6, 7, 3\}} = 2$$



Moat-Growing Process

$$S = \{s, 2, 6, 7, 3, 5\}$$

$$T = \{(s, 2), (s, 6), (s, 7), (6, 3), (3, 5)\}$$

$$y_{\{s\}} = 9$$

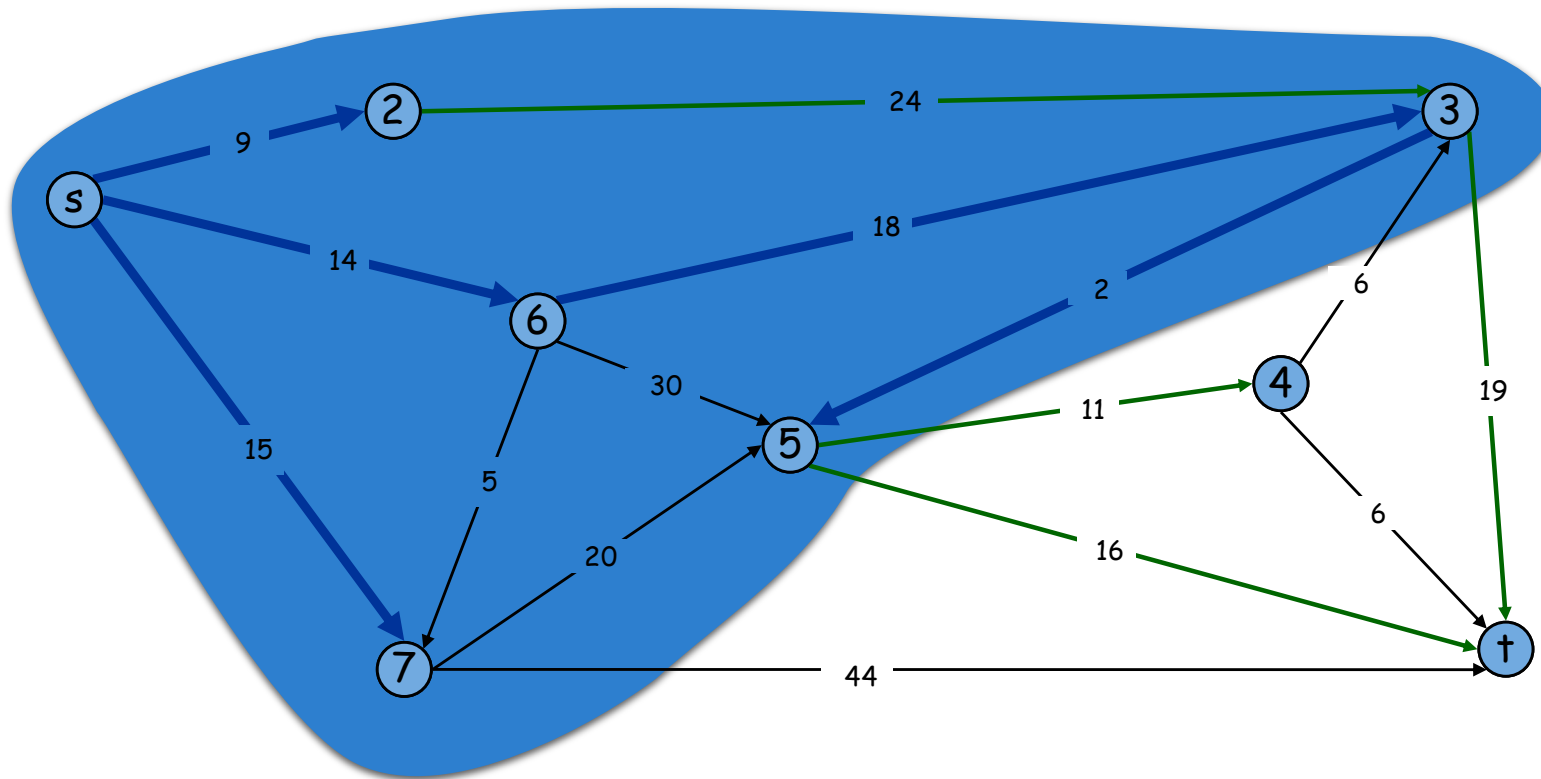
$$y_{\{s, 2\}} = 5$$

$$y_{\{s, 2, 6\}} = 1$$

$$y_{\{s, 2, 6, 7\}} = 17$$

$$y_{\{s, 2, 6, 7, 3\}} = 2$$

$$y_{\{s, 2, 6, 7, 3, 5\}} = ?$$



Moat-Growing Process

$$S = \{s, 2, 6, 7, 3, 5\}$$

$$T = \{(s, 2), (s, 6), (s, 7), (6, 3), (3, 5), (5, 4)\}$$

$$y_{\{s\}} = 9$$

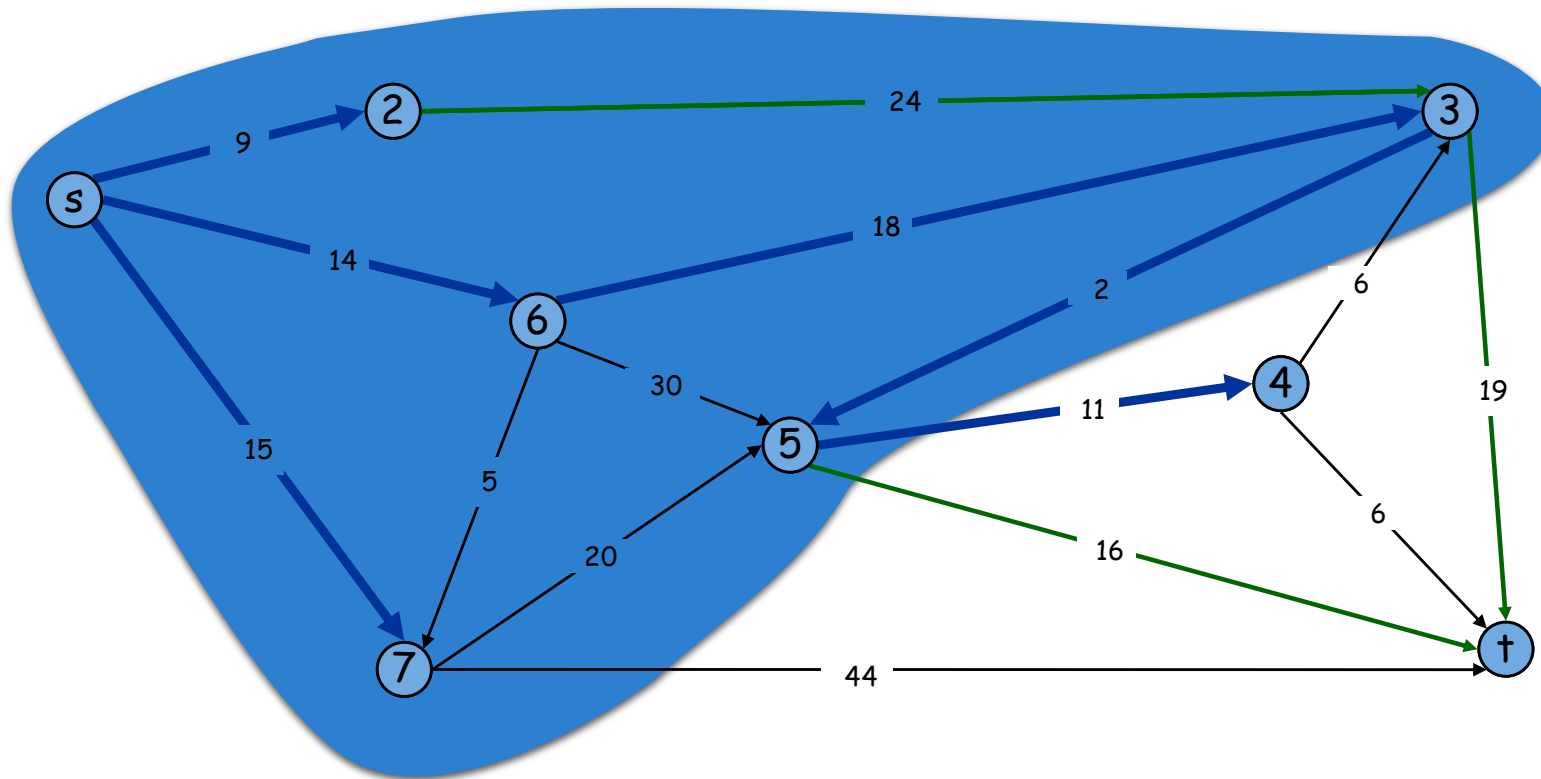
$$y_{\{s, 2\}} = 5$$

$$y_{\{s, 2, 6\}} = 1$$

$$y_{\{s, 2, 6, 7\}} = 17$$

$$y_{\{s, 2, 6, 7, 3\}} = 2$$

$$y_{\{s, 2, 6, 7, 3, 5\}} = 11$$



Moat-Growing Process

$S = \{s, 2, 6, 7, 3, 5, 4\}$

$T = \{(s, 2), (s, 6), (s, 7), (6, 3), (3, 5), (5, 4)\}$

$$y_{\{s\}} = 9$$

$$y_{\{s, 2\}} = 5$$

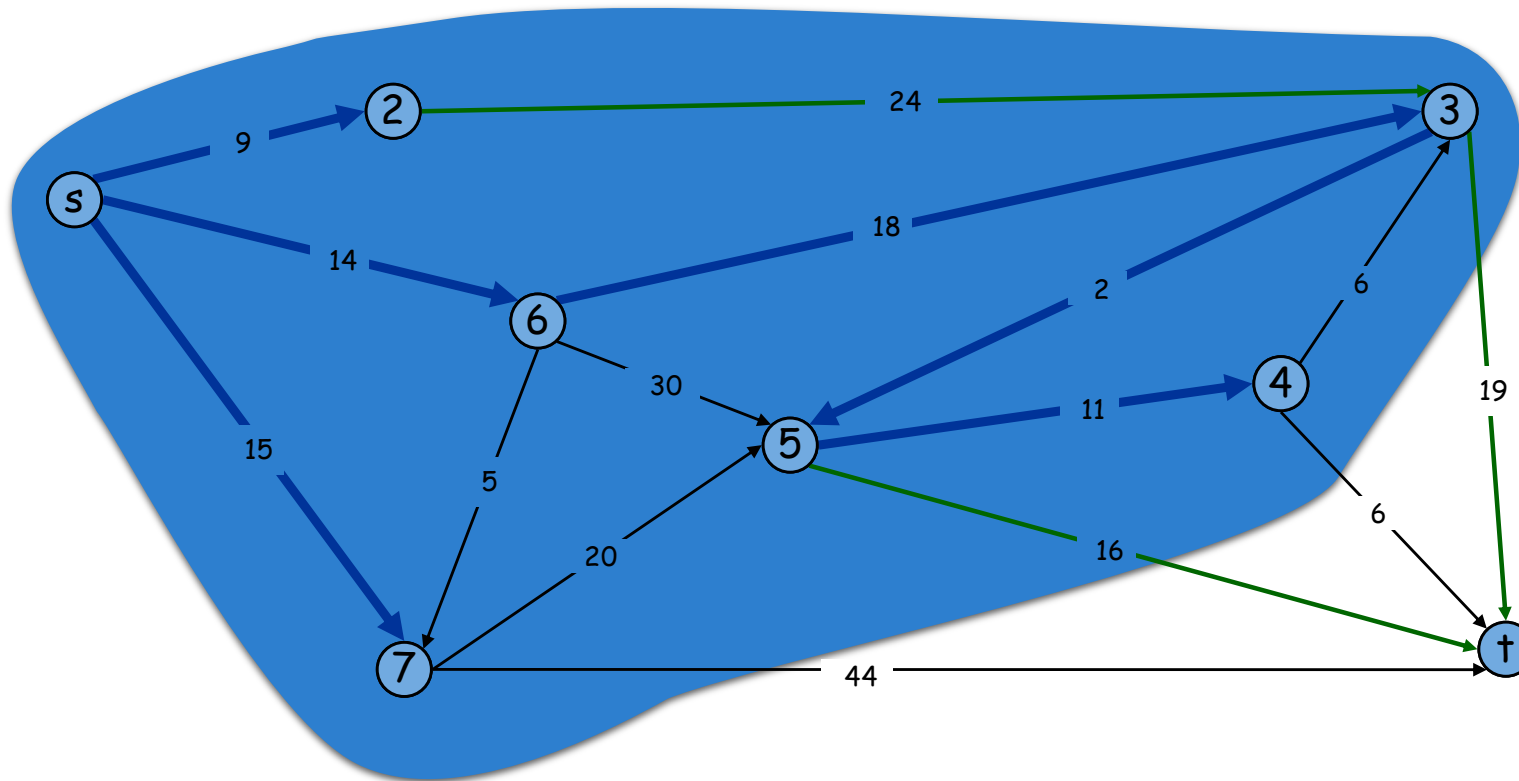
$$y_{\{s, 2, 6\}} = 1$$

$$y_{\{s, 2, 6, 7\}} = 17$$

$$y_{\{s, 2, 6, 7, 3\}} = 2$$

$$y_{\{s, 2, 6, 7, 3, 5\}} = 11$$

$$y_{\{s, 2, 6, 7, 3, 5, 4\}} = ?$$



Moat-Growing Process

$$S = \{s, 2, 6, 7, 3, 5, 4\}$$

$$T = \{(s, 2), (s, 6), (s, 7), (6, 3), (3, 5), (5, 4)\}$$

$$y_{\{s\}} = 9$$

$$y_{\{s, 2\}} = 5$$

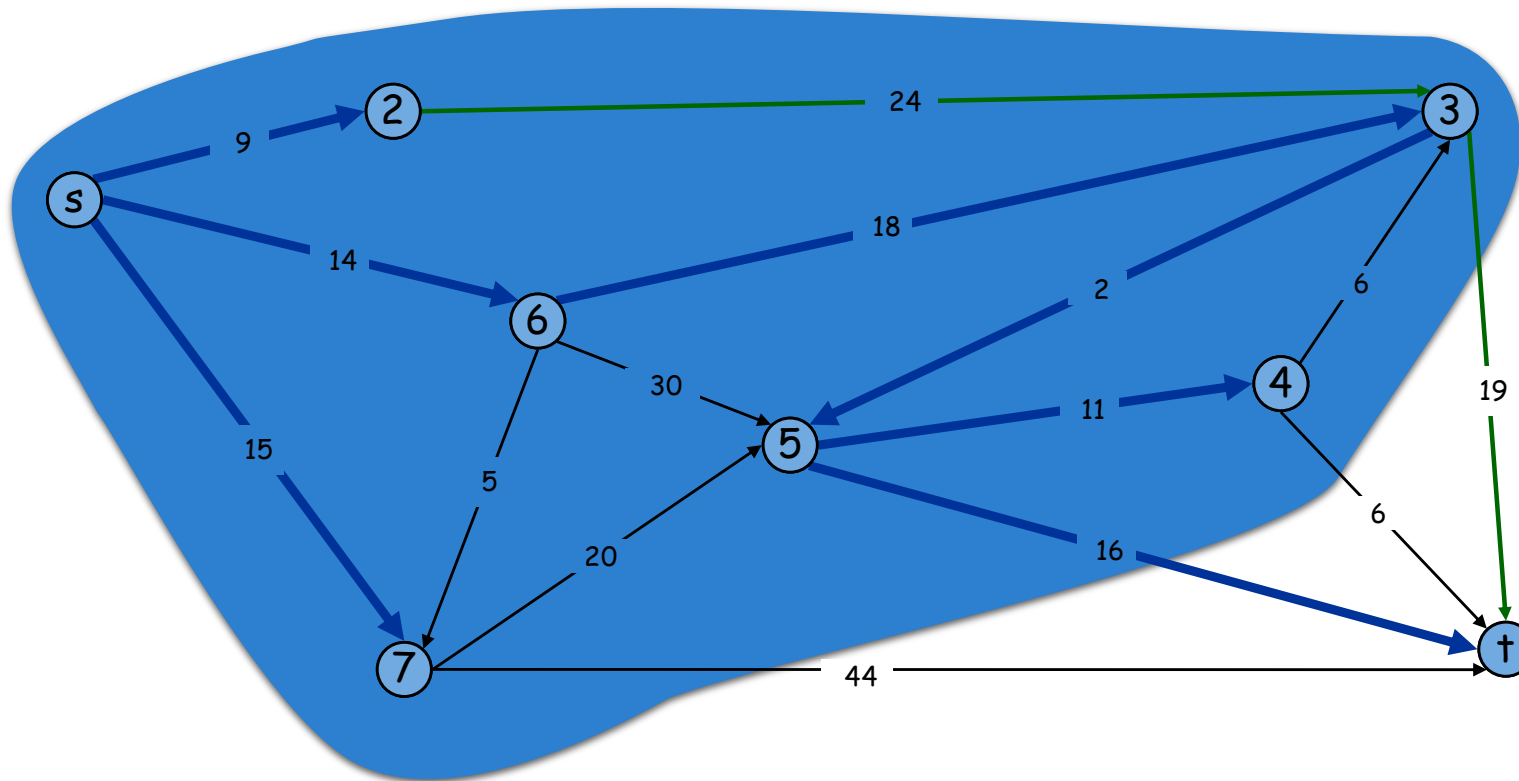
$$y_{\{s, 2, 6\}} = 1$$

$$y_{\{s, 2, 6, 7\}} = 17$$

$$y_{\{s, 2, 6, 7, 3\}} = 2$$

$$y_{\{s, 2, 6, 7, 3, 5\}} = 11$$

$$y_{\{s, 2, 6, 7, 3, 5, 4\}} = 5$$

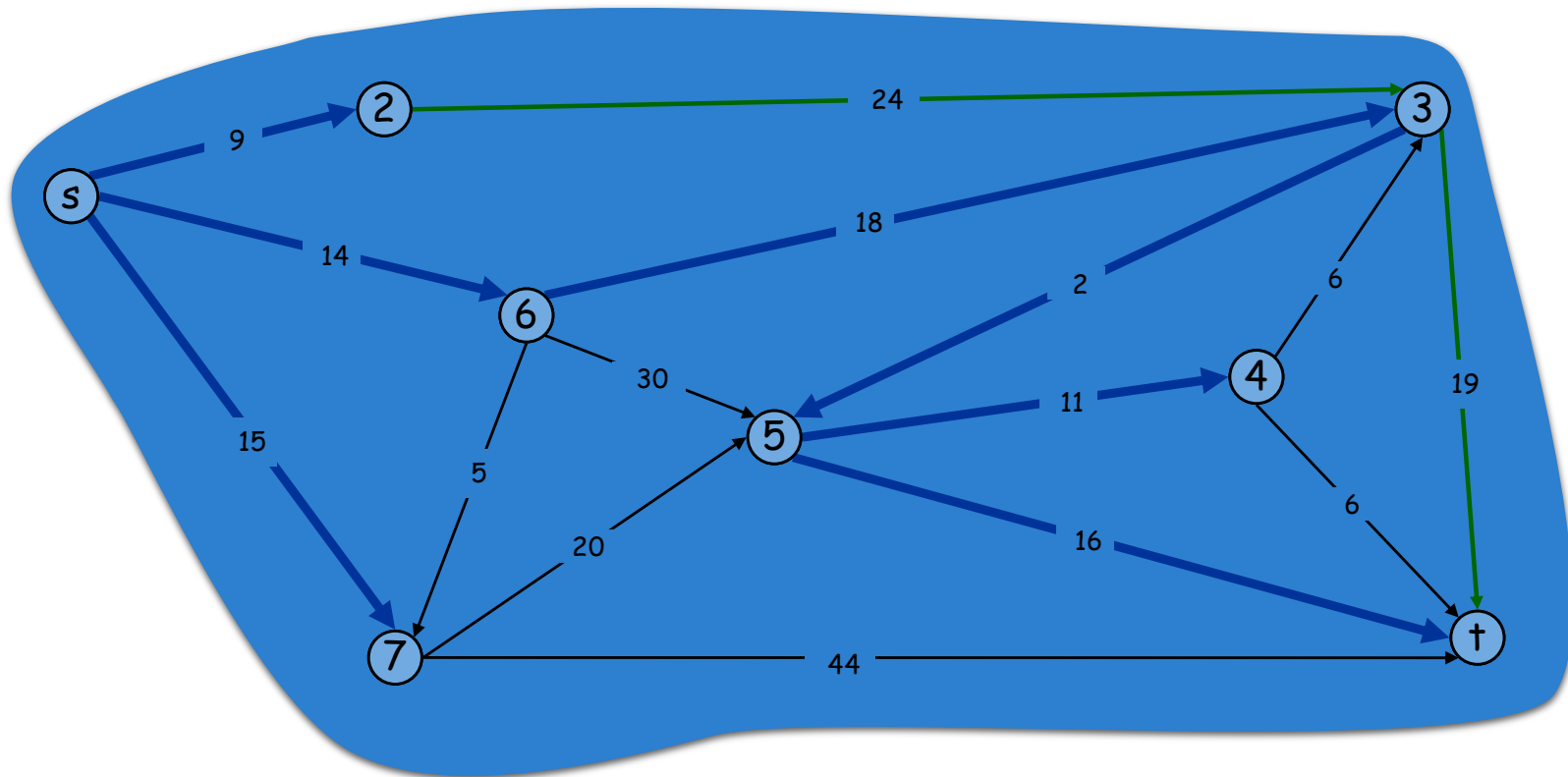


Moat-Growing Process

$S = \{s, 2, 6, 7, 3, 5, 4, 5\}$
 $T = \{(s, 2), (s, 6), (s, 7), (6, 3), (3, 5), (5, 4), (5, t)\}$

$$\begin{aligned}
 y_{\{s\}} &= 9 \\
 y_{\{s, 2\}} &= 5 \\
 y_{\{s, 2, 6\}} &= 1 \\
 y_{\{s, 2, 6, 7\}} &= 17
 \end{aligned}$$

$$\begin{aligned}
 y_{\{s, 2, 6, 7, 3\}} &= 2 \\
 y_{\{s, 2, 6, 7, 3, 5\}} &= 11 \\
 y_{\{s, 2, 6, 7, 3, 5, 4\}} &= 5
 \end{aligned}$$



Moat-Growing Process

$S = \{s, 2, 6, 7, 3, 5, 4, 5\}$

$T = \{(s, 2), (s, 6), (s, 7), (6, 3), (3, 5), (5, 4), (5, t)\}$

$$y_{\{s\}} = 9$$

$$y_{\{s, 2\}} = 5$$

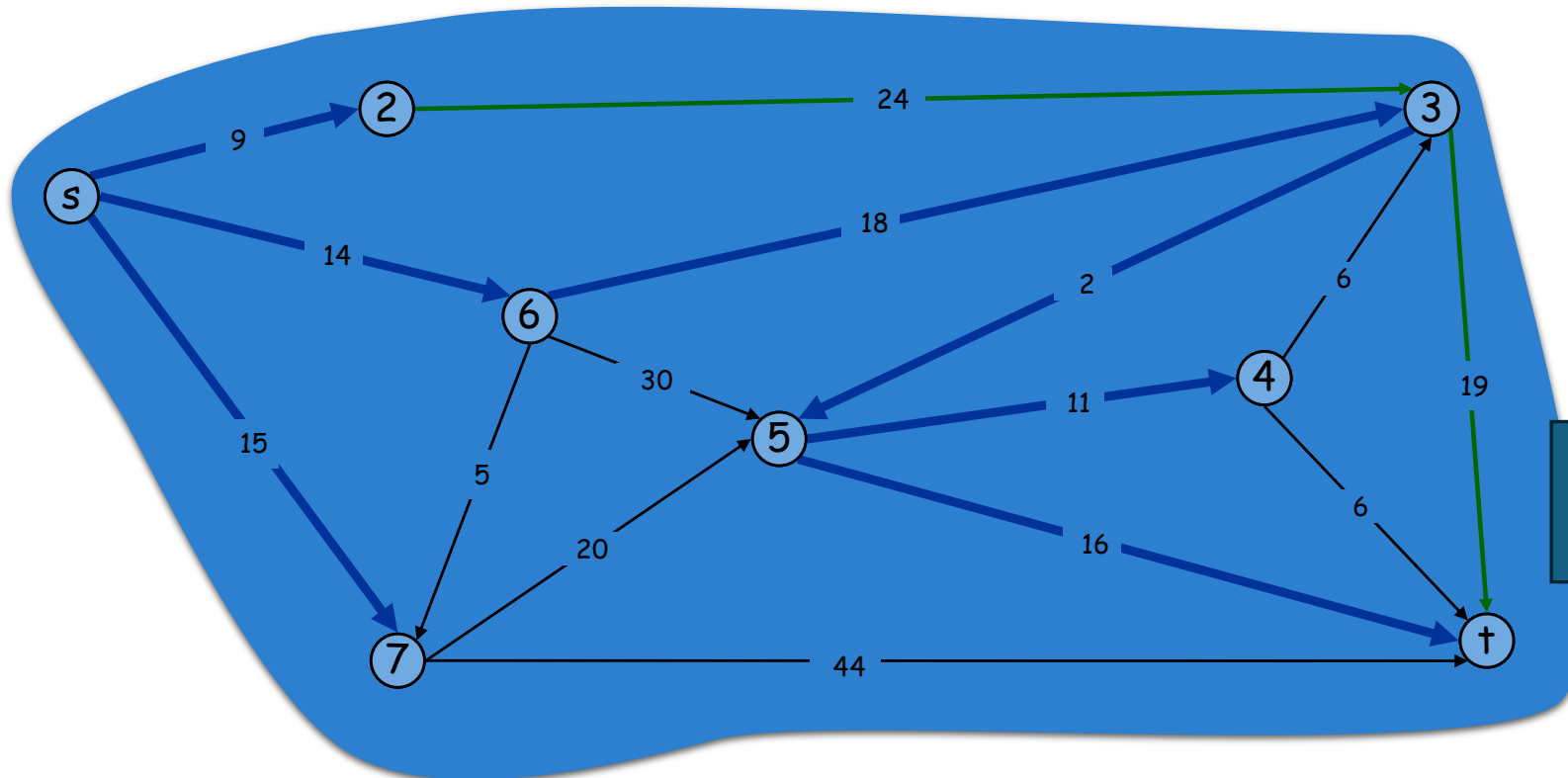
$$y_{\{s, 2, 6\}} = 1$$

$$y_{\{s, 2, 6, 7\}} = 17$$

$$y_{\{s, 2, 6, 7, 3\}} = 2$$

$$y_{\{s, 2, 6, 7, 3, 5\}} = 11$$

$$y_{\{s, 2, 6, 7, 3, 5, 4\}} = 5$$



Shortest Path
 $14 + 18 + 2 + 16 = 50$

Dual
 $9 + 5 + 1 + 17 + 2 + 11 + 5 = 50$



Primal-Dual Algorithm: Moat-Growing

- Initialize $T = \emptyset$, $y = 0$, $S = \{s\}$
- While $t \notin S$:
 - Raise y_S until some edge (u, v) tight
 - Add (u, v) to T
- Output (s, t) -path P in T

Analysis

- (Primal CS) If $e \in P$, then $\sum_{e \in \delta(S)} y_S = c_e$ ✓
- (Dual CS) If $y_S > 0$, then $|P \cap \delta(S)| = 1$

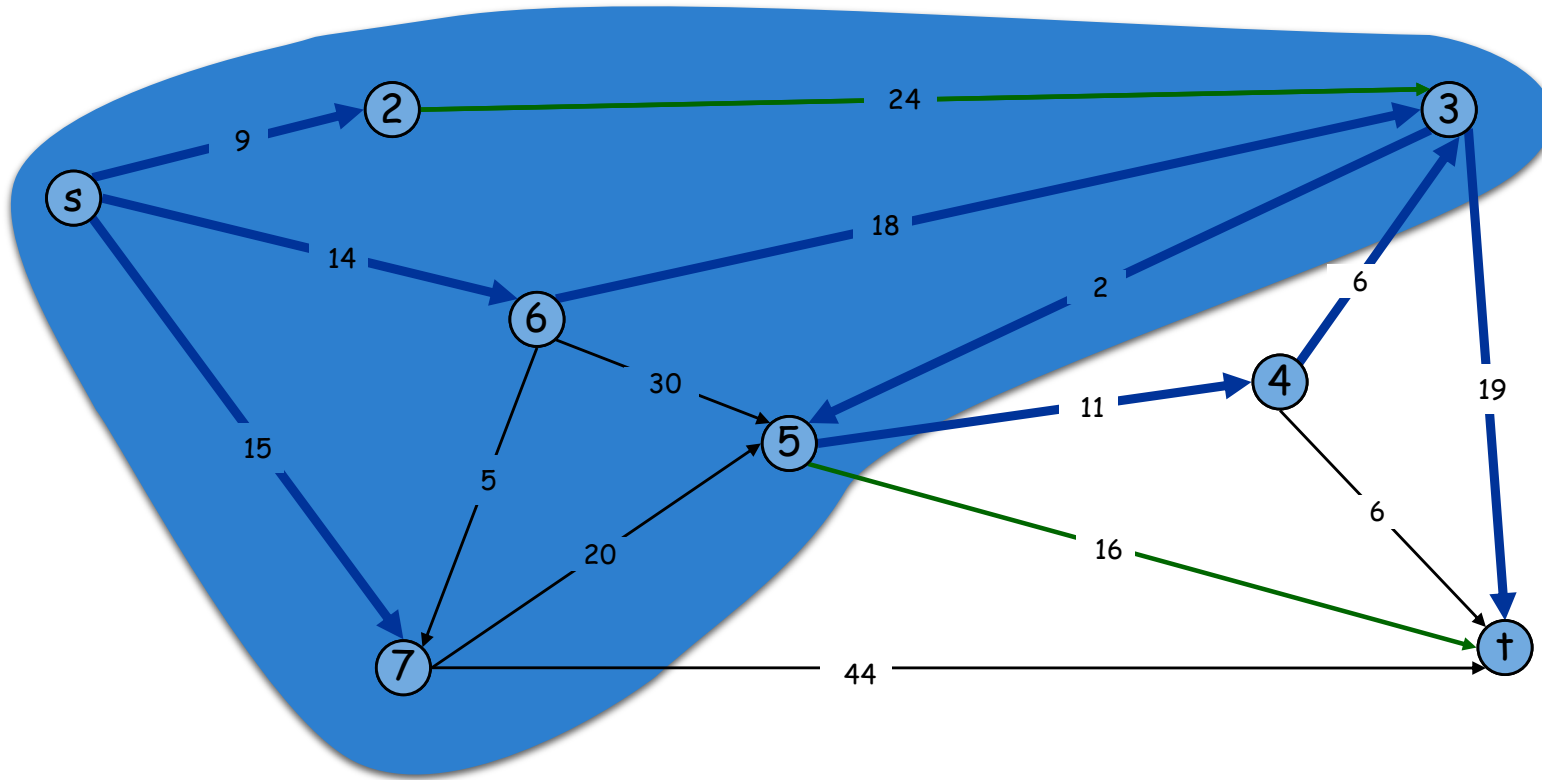
$$\begin{aligned}
 &\text{minimize} && \sum_{e \in E} c_e x_e \\
 &\text{subject to} && \sum_{e \in \delta(S)} x_e \geq 1 \quad \forall S \in \mathcal{C} \\
 &&& x \geq 0
 \end{aligned}$$

$$\begin{aligned}
 &\text{maximize} && \sum_{S \in \mathcal{C}} y_S \\
 &\text{subject to} && \sum_{S: e \in \delta(S)} y_S \leq c_e \quad \forall e \in E \\
 &&& y \geq 0
 \end{aligned}$$

Intuition: Dual = “moat” packing

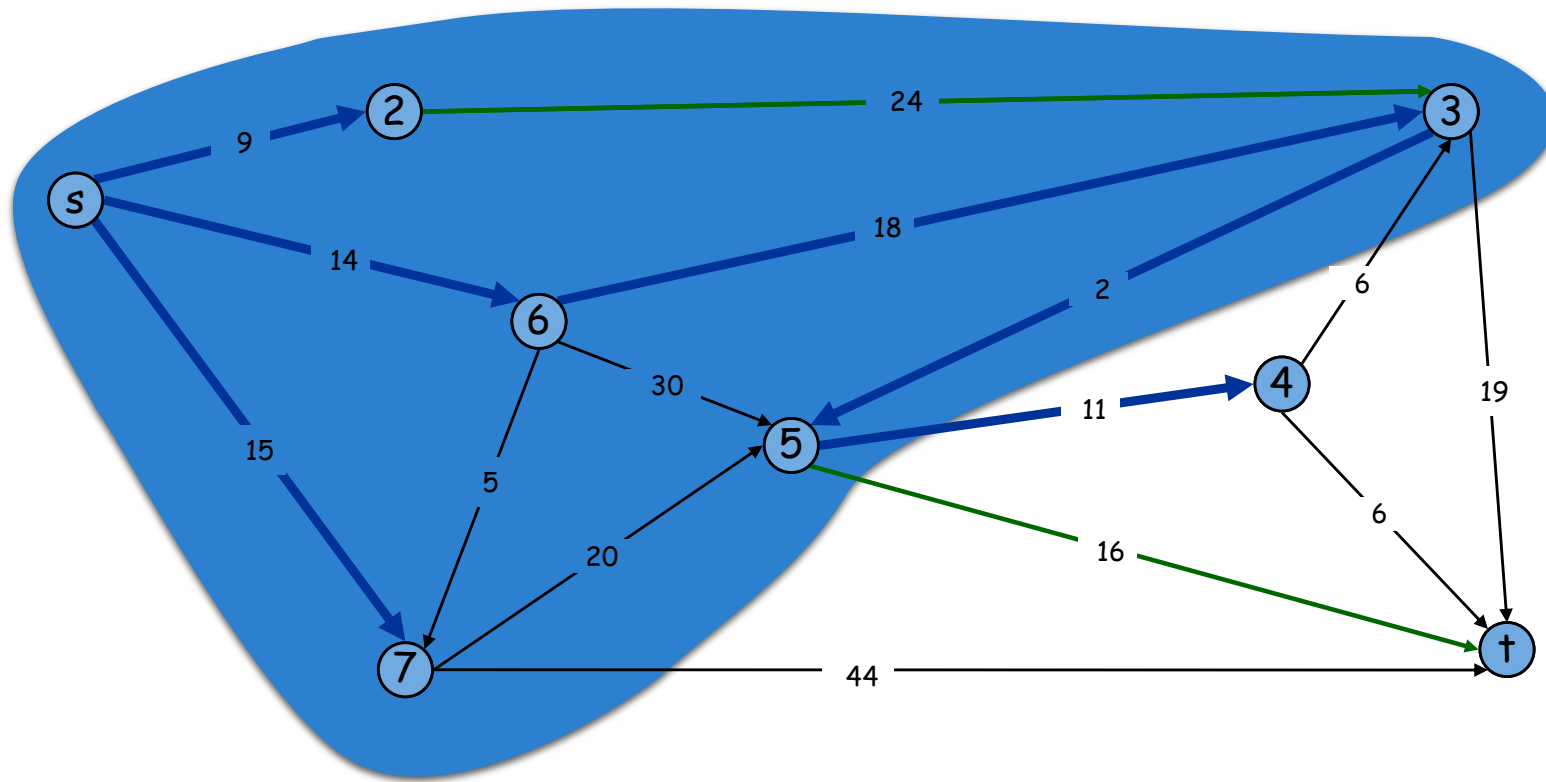
Lem. If $y_s > 0$, then $|P \cap \delta(S)| = 1$

Proof: If $|P \cap \delta(S)| > 1$, then P left S , re-entered and left again



Lem. If $y_s > 0$, then $|P \cap \delta(S)| = 1$

Proof: If $|P \cap \delta(S)| > 1$, then P left S , re-entered and left again



When $S = \{s, 2, 6, 7, 3, 5\}$,

- Added $(5, 4)$
- T is a tree spanning S

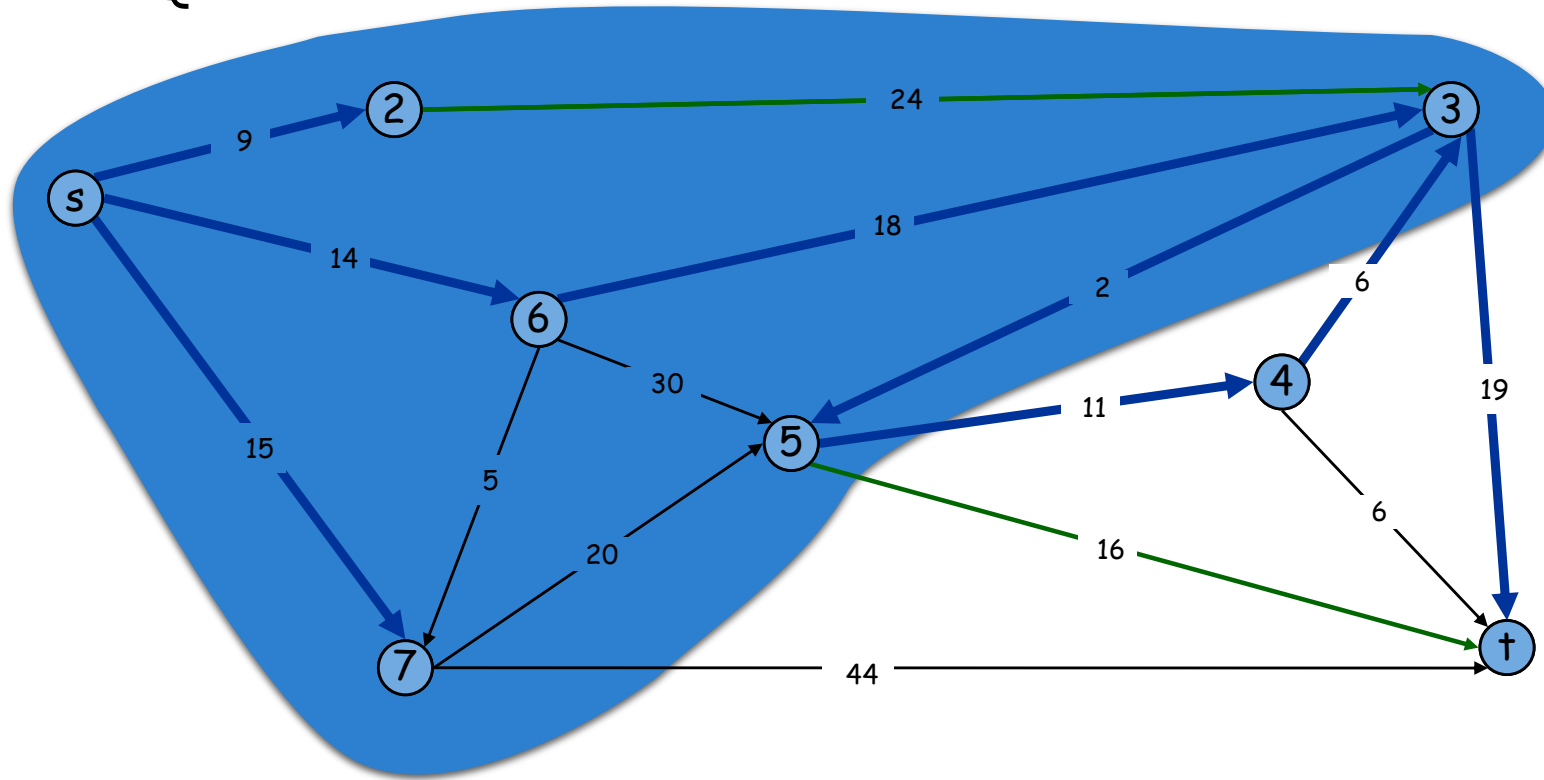
$(4, 3)$ and $(3, t)$ added later

Lem. If $y_s > 0$, then $|P \cap \delta(S)| = 1$

Proof: If $|P \cap \delta(S)| > 1$, then P left S , re-entered and left again

Implies a cycle in T at some point, a contradiction.

QED



When $S = \{s, 2, 6, 7, 3, 5\}$,

- Added (5,4)
- T is a tree spanning S

(4,3) and (3,t) added later

Primal-Dual Algorithm: Moat-Growing

- Initialize $T = \emptyset$, $y = 0$, $S = \{s\}$
- While $t \notin S$:
 - Raise y_S until some edge (u, v) tight
 - Add (u, v) to T
- Output (s, t) -path P in T

Analysis

- (Primal CS) If $e \in P$, then $\sum_{e \in \delta(S)} y_S = c_e$ ✓
- (Dual CS) If $y_S > 0$, then $|P \cap \delta(S)| = 1$ ✓
- Conclude that P is shortest path

$$\begin{aligned}
 &\text{minimize} && \sum_{e \in E} c_e x_e \\
 &\text{subject to} && \sum_{e \in \delta(S)} x_e \geq 1 \quad \forall S \in \mathcal{C} \\
 &&& x \geq 0
 \end{aligned}$$

$$\begin{aligned}
 &\text{maximize} && \sum_{S \in \mathcal{C}} y_S \\
 &\text{subject to} && \sum_{S: e \in \delta(S)} y_S \leq c_e \quad \forall e \in E \\
 &&& y \geq 0
 \end{aligned}$$

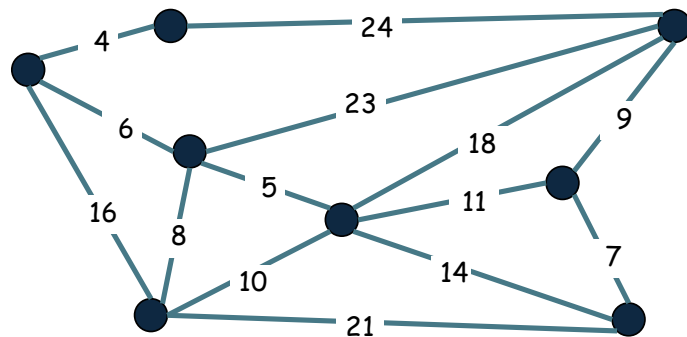
Minimum Spanning Trees

$$x_e$$

$$\forall u, v \in V \quad \sum_{e \in \text{path}(u,v)} x_e \geq 1$$

$$\sum_{e \in \delta(S)} x_e \geq 1 \quad \forall \emptyset \subsetneq S \subsetneq V$$

- Given undirected graph $G = (V, E)$ with edge costs c_e , find MST
- How to write LP?
- Cut constraints!

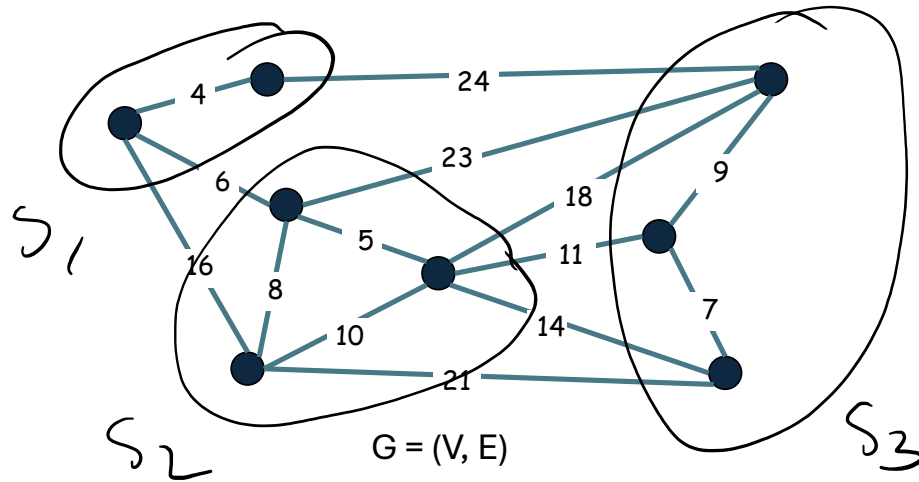


$$\begin{aligned} &\text{minimize} && \sum_{e \in E} c_e x_e \\ &\text{subject to} && \sum_{S: e \in \delta(S)} x_e \geq 1 \quad \forall \emptyset \subsetneq S \subsetneq V \\ &&& x \geq 0 \end{aligned}$$

- Turns out can only prove 2-approximation ☹️

Partition Constraints

- Vertex partition $\Pi = (S_1, \dots, S_k)$, parts S_i .
- Let $\delta(\Pi) = \{(u, v) : u \in S_i, v \in S_j, i \neq j\}$, edges crossing parts
- Spanning tree T has $|T \cap \delta(\Pi)| \geq |\Pi| - 1$

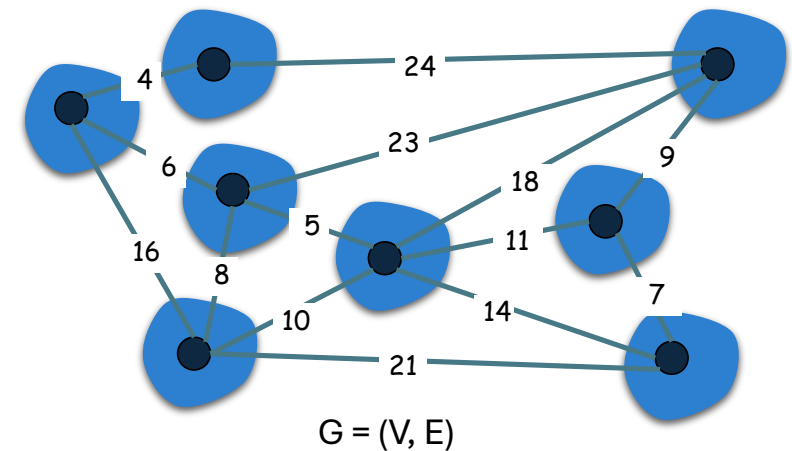


$$\begin{aligned}
 &\text{minimize} && \sum_{e \in E} c_e x_e \\
 &\text{subject to} && \sum_{\Pi: e \in \delta(\Pi)} x_e \geq |\Pi| - 1 \quad \forall \Pi \\
 &&& x \geq 0
 \end{aligned}$$

$$\begin{aligned}
 &\text{maximize} && \sum_{\Pi} (|\Pi| - 1) y_{\Pi} \\
 &\text{subject to} && \sum_{\Pi: e \in \delta(\Pi)} y_{\Pi} \leq c_e \quad \forall e \in E \\
 &&& y \geq 0
 \end{aligned}$$

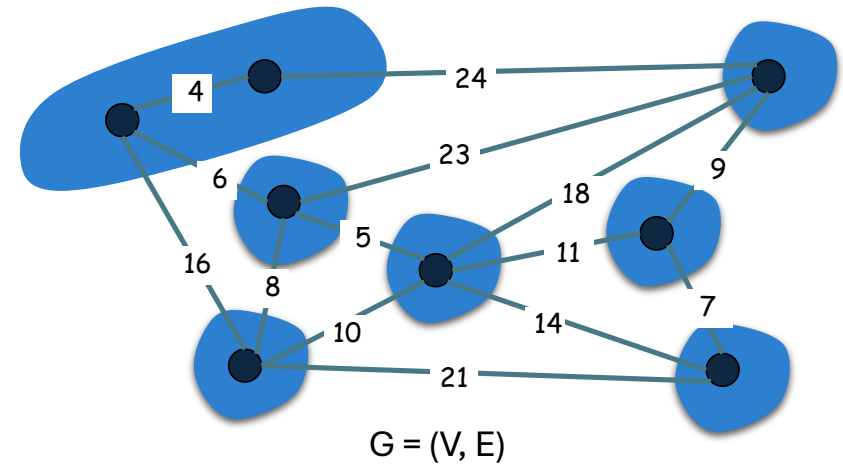
Primal-Dual Algorithm: Moat-Growing

- Initialize $T = \emptyset$, $y = 0$, $\Pi = (\{v\}_{v \in V})$ (i.e. singletons)
- While $\Pi \neq (V)$:
 - Raise y_Π until some edge (u, v) tight
 - Add (u, v) to T
 - Merge parts
- Output T



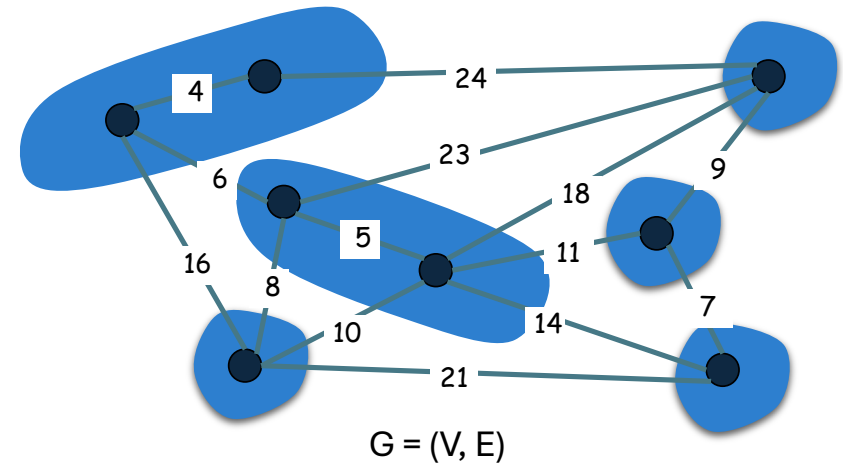
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- Initialize $T = \emptyset$, $y = 0$, $\Pi = (\{v\}_{v \in V})$ (i.e. singletons)
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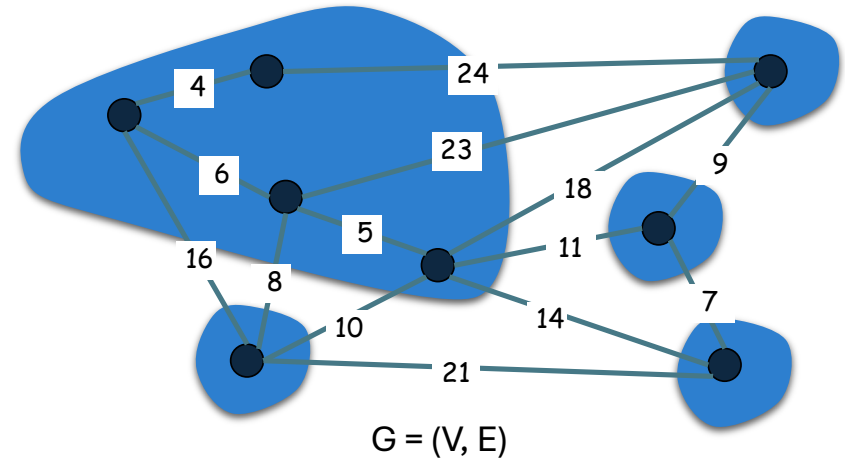
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 - Raise y_Π until some edge (u, v) tight
 - Add (u, v) to T
 - Merge parts
- Output T



Analysis

- (Primal CS) If $e \in T$, then $\sum_{e \in \delta(\Pi)} y_\Pi = c_e$
- (Dual CS) If $y_\Pi > 0$, then $|T \cap \delta(\Pi)| = |\Pi| - 1$
 - When we are raising y_Π , Π is the connected components of T
- Conclude T is MST

Basic Primal-Dual for Covering Problems

Primal-Dual Algorithm:

- Initialize $X, y = 0$
- While some covering constraint violated,
 - raise its dual variable until dual constraint tight
 - Set its primal variable to 1

Analysis:

- (Primal CS) If $X_i = 1$, then $\sum_{e \in S_i} y_e = w_i$
- (α -approx. Dual CS) If $y_e > 0$, then $\sum_{i: S_i \ni e} X_i \leq \alpha$
- Conclude algorithm is α -approximation
- 🎉

$$\begin{array}{ll}\text{minimize} & \sum_{1 \leq i \leq n} c_i x_i \\ \text{subject to} & \sum_{1 \leq i \leq n} b_{ij} x_i \geq d_j \quad \forall 1 \leq j \leq m \\ & x \geq 0\end{array}$$

$$\begin{array}{ll}\text{maximize} & \sum_{1 \leq j \leq n} d_j y_j \\ \text{subject to} & \sum_{1 \leq j \leq n} b_{ij} y_j \leq c_i \quad \forall 1 \leq i \leq n \\ & y \geq 0\end{array}$$